

Optimal allocations with capacity constrained verification*

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Abstract

A principal has m identical objects to allocate among a group of n agents. Objects are desirable and the principal's value of assigning an object to a given agent is that agent's private information. The principal can verify up to k agents, where $k < m$, thereby perfectly learning their types. We show that when no monetary transfers are available, optimal mechanisms take a merit-with-guarantee form. Such mechanisms can be implemented in two stages: in the first stage, objects are allocated based on merit, with up to k agents verified; in the second stage, any remaining objects are distributed via a lottery that guarantees all low-reporting agents a positive allocation probability. This guarantee sustains truthful reporting and allows the mechanism to allocate efficiently to more agents than can be verified. Truthful reporting is a Bayes–Nash equilibrium but not an ex post equilibrium.

Keywords: Mechanism Design with Evidence; Allocations; Verification

JEL classification: D82, D71

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1 Introduction

We examine an allocation problem with multiple identical objects. Agents possess private information about the principal’s value of assigning them an object. Each agent desires at most one object, and monetary transfers are not available. Agents’ private information is based on verifiable evidence, but the principal is constrained to verify only a fixed number of reports—fewer than the number of available objects. We study how the principal should optimally audit agents based on their reports and use the audit outcomes, together with agents’ reports, to allocate objects. This problem differs from standard mechanism design due to the verifiability of private information, which arises in various economic contexts.

Many allocation problems share this structure. For example, public agencies often allocate goods on the basis of recipients’ need. Such goods include social housing, energy subsidies, or other social benefits. Applicants’ need levels—based on income, household composition, or other characteristics—are private but verifiable through documentation, consultations, or home visits. However, not all recipients can be verified, and many households necessarily receive benefits without thorough verification of their claims. A similar structure arises when goods are allocated on the basis of merit, such as research grants or work permits for skilled immigrants. Many countries operate points-based immigration systems where applicants’ qualifications determine their eligibility for work permits or permanent residence (Miller 1999, Sweetman and Warman 2013). In these settings, more objects must be allocated than can be verified, and the principal faces a hard capacity constraint—a fixed number of caseworkers, auditors, or officers—rather than facing a marginal cost of verification.

To illustrate the environment we have in mind, consider the allocation of social housing in England. Local authorities maintain housing registers and allocate homes using a points-based system that ranks applicants by housing need, based on self-reported characteristics such as overcrowding, medical conditions, or risk of homelessness. Each application is assessed by a caseworker through desk-based checks, supporting documentation, and in some cases home visits. However, with long waiting lists and a limited number of caseworkers, not all claims can be thoroughly verified before allocation decisions are made. Among those who receive housing, only a subset are selected for detailed verification, with some cases selected at random for review. The setting thus exhibits the key features of our model: objects are allocated on the basis of privately known but verifiable characteristics, the number of objects allocated exceeds the number of verifications that can be conducted, and audits are conducted randomly among recipients. We analyze the optimal method for allocating m homes among n agents when at most $k < m$ agents can be verified.

To study this problem, we formulate a model of capacity-constrained verification in an allocation problem without transfers. We depart from the standard assumption that

verification is costly at the margin (Townsend 1979, Gale and Hellwig 1985, Ben-Porath, Dekel and Lipman 2014) and instead assume that the principal faces a fixed verification capacity. In the applications we have in mind, verification requires specialized human capital whose supply is fixed in the short run: caseworkers qualified to carry out home visits, trained immigration officers who can authenticate foreign credentials and conduct interviews, or academic experts with the narrow expertise needed to evaluate research proposals. Expanding this capacity requires recruiting and training new personnel, a process that operates on a much longer time scale than the time available before the allocation has to be finalized. A local housing authority assigning social housing cannot easily expand its team of caseworkers to match a larger pool of applicants with greater needs; a migration agency deciding how to allocate work permits this month cannot hire additional officers in response to a surge in high-quality applications; and a funding body reviewing grant applications cannot enlarge its expert panel after the call has closed. In these settings, verification capacity is effectively predetermined at the time the allocation decision is made, and the relevant constraint is not the marginal cost of an additional audit but the total number of audits that can be conducted.

Before allocating objects, the principal may audit a subset of agents and learn their types perfectly. At most k agents can be audited. The principal obtains value $v(t_i)$ from assigning an object to agent i , and total payoff is the sum of values $v(t_i)$ across all assigned agents i . There are no monetary transfers and no direct costs of verification; the only constraint is the capacity limit on the number of checks. The mechanism design question is whom to check based on the submitted reports and how to allocate objects based on the reports and checks.

We find mechanisms that maximize the expected payoff from allocating m objects subject to k checks. The allocation rules in the optimal mechanisms can be decomposed into two rules that operate in two stages: a merit-based rule in the first stage and a lottery-based rule in the second stage. The merit-based rule allocates objects “efficiently” to agents with the highest reported types, where the number of objects allocated in this way is governed by two or three thresholds. With two thresholds, if fewer than k agents report a type above the lower threshold, all such agents receive an object and the remaining objects enter the second stage. If more than k agents report a type above the lower threshold, an agent receives an object in the first stage if either (i) his report is among the k highest, or (ii) his report is among the m highest and exceeds the higher threshold. Among agents who receive an object in the first stage, k are randomly selected for verification, and any agent found lying receives no object. Agents who do not receive an object in the first stage and are not caught lying enter the second stage, where a lottery-based rule allocates the remaining objects so that all types below the lower threshold receive an object with the same probability. If there is a third threshold, the allocation is similar but an agent also obtains an object if his report is above the lowest threshold and

among the m highest reports.

To better understand the optimal mechanism, we discuss incentives for truthful reporting, note that the merit-based rule sometimes allocates more than k objects, so not all agents receiving an object in the first stage can be checked. This suggests that agents might be tempted to claim that they have a high type. The optimal mechanism mitigates these incentives by not always allocating all objects in the first stage: the lottery-based allocation of remaining objects ensures that even agents with low types retain a positive probability of receiving an object when reporting truthfully. In particular, their expected probability of obtaining an object through the lottery equals the expected probability of obtaining an object in the first stage by reporting a lie. However, if a low-type agent knew that more than m agents have types above the higher threshold, he would anticipate that all objects will be allocated in the first stage and that truthful reporting yields zero probability of receiving an object, since there are no objects left for the lottery stage. In that case, he would have a strict incentive to misreport to a high type and hope not to be verified. Thus, while the optimal mechanism is Bayesian incentive compatible, it does not satisfy the stronger notion of ex-post incentive compatibility. This contrasts with models of costly verification, where optimal mechanisms can typically be implemented to satisfy these stronger incentive constraints.

1.1 Related Literature

The literature on costly state verification in principal-agent models was initiated by Townsend (1979). Our model differs from his, and the literature building on it (see e.g. Gale and Hellwig 1985, Border and Sobel 1987), since monetary transfers are not feasible in our model. In models without transfers, incentive constraints and economic trade-offs are different than in models with money.

Over the last years, there has been growing interest in models with state verification that do not allow for transfers, as initiated by Ben-Porath et al. (2014). The model studied by Ben-Porath et al. consists of a principal who wishes to allocate a single indivisible good among a group of agents, and each agent's type can be learned at a given cost. They find that the expected utility-maximizing mechanism for the principal is a favored-agent mechanism, where a pre-determined favored agent receives the object unless another agent claims a value above a threshold, in which case the agent with the highest (net) type gets the object. Our model is similar in that there are no monetary transfers but differs in that our principal allocates several objects and can check a fixed number of agents at no cost. Compared to a model with a possibly unlimited number of tests at a constant marginal cost, this leads to a richer set of trade-offs that determine optimal mechanisms.

Ben-Porath, Dekel and Lipman (2019) analyze optimal mechanisms in a model of Dye evidence and apply their results to a model in which the principal allocates m objects

and can verify agents at a constant marginal cost. They show that the optimal mechanism generalizes the favored-agent mechanism in a natural way to an m -favored-agents mechanism. Their approach implies that the optimal mechanism satisfies stronger ex-post incentive constraints, which shows a conceptual difference to our model, where the optimal Bayesian incentive-compatible mechanism is not ex post incentive compatible. This also suggests that the optimal mechanism in our model cannot be obtained using their general results for the Dye-evidence model. Other related papers on mechanism design with evidence include Glazer and Rubinstein (2004) and Glazer and Rubinstein (2006), which consider a situation in which an agent has private information about several characteristics and tries to persuade a principal to take a given action, and the principal can only check one of the agent’s characteristics.¹ Other papers on optimal mechanisms in settings with evidence include Mylovanov and Zapechelnyuk (2017), Halac and Yared (2020), and Kattwinkel and Knoepfle (2023).

Methodologically, we build on results showing for which interim rules there exist corresponding ex post rules. This approach was pioneered by Maskin and Riley (1984), who used it to characterize optimal auctions with risk-averse bidders. This question was first answered conclusively by Border (1991), who characterized the set of feasible interim allocation rules. More recently, Border’s result was generalized by Che, Kim and Mierendorff (2013), which, among other things, allow for the allocation of m objects when agents face capacity constraints. We use a recent generalization by Valenzuela-Stookey (2024) that extends the characterization of feasible interim rules to settings with type-dependent ex-post constraints and that perfectly fits our setting.

The rest of the paper is organized as follows. In Section 2, we present the model, and characterize incentive-compatible mechanisms. In Section 3, we describe the class of optimal mechanisms and discuss its main qualitative properties. In Section 4, we provide an argument for why it is optimal. In Section 5, we show how to determine the expected probability of winning at the lottery stage in the optimal mechanism and analyze the mechanism design problem for a large market. Section 6 concludes.

2 Model and preliminaries

A principal allocates m identical and indivisible objects to a set $N := \{1, \dots, n\}$ of agents, where $m < n$. Each agent i is characterized by a type $t \in T := [0, 1]$, which is private information to agent i . If agent i with type t obtains an object, agent i ’s utility is $u_i(t) > 0$ and the principal’s utility from allocating to i is $v(t)$, where v is strictly increasing and

¹For additional papers on mechanism design with evidence, see also Green and Laffont (1986), Bull and Watson (2007), Deneckere and Severinov (2008), Ben-Porath and Lipman (2012), and Schweighofer-Kodritsch and Strausz (2024).

satisfies $v(0) = 0$. No agent needs more than one object. Types are drawn independently from a uniform distribution and we denote $\mathbf{t} = (t_i)_{i \in N}$, $\mathbf{t}_{-i} = (t_j)_{j \in N \setminus \{i\}}$, and $\mathbf{T} = [0, 1]^N$.² The principal has a verification technology at his disposal and can learn the types of up to $k < m$ agents perfectly (we will use verification and audit interchangeably). The principal can commit to arbitrary mechanisms to allocate the objects, and we consider Bayes-Nash equilibria of the induced game.

We now define direct mechanisms for this setting. We capture stochastic mechanisms by using a random variable s , which is distributed independently of the types and uniformly on $[0, 1]$ and whose realization is only observed by the principal. Formally, an *allocation rule* is a profile of functions $p_i : \mathbf{T} \times [0, 1] \rightarrow \{0, 1\}$ satisfying, for all $\mathbf{t}$ and s , $\sum_{i=1}^n p_i(\mathbf{t}, s) \leq m$ and an *audit rule* is a profile of functions $a_i : \mathbf{T} \times [0, 1] \rightarrow \{0, 1\}$ satisfying, for all $\mathbf{t}$ and s , $\sum_{i=1}^n a_i(\mathbf{t}, s) \leq k$.³

A *direct mechanism* (p, a) consists of an allocation rule and an audit rule. Given a profile of reports $\mathbf{t}$ and a realization s of the random variable, agent i is audited if $a_i(t_i, \mathbf{t}_{-i}, s) = 1$. We say that a *lie by agent i is detected* if i 's true type t'_i differs from his report t_i and $a_i(t_i, \mathbf{t}_{-i}, s) = 1$. Agent i obtains an object if no lie by agent i was detected and $p_i(t_i, \mathbf{t}_{-i}, s) = 1$, and obtains no object otherwise.

One can show that it is without loss of generality to restrict attention to direct mechanisms that are Bayesian incentive-compatible.⁴

Definition 1. A mechanism (p, a) is *Bayesian incentive compatible (BIC)* if, for all $i \in N$ and all $t, t' \in T$

$$u_i(t') \cdot \mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t', \mathbf{t}_{-i}, s)] \geq u_i(t') \cdot \mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t, \mathbf{t}_{-i}, s)(1 - a_i(t, \mathbf{t}_{-i}, s))]. \quad (1)$$

The principal's problem can formally be stated as:

$$\begin{aligned} \max_{p, a} \mathbb{E}_{\mathbf{t}, s} \left[\sum_{i \in N} p_i(t_i, \mathbf{t}_{-i}, s) v(t_i) \right] \\ \text{s.t. } (p, a) \text{ is BIC,} \end{aligned}$$

and we say a mechanism is *optimal* if it solves this problem.

Reformulating the principal's problem By symmetry, it is without loss of optimality to restrict attention to symmetric mechanisms that treat all agents identically, and

²More generally, the i.i.d. case with a continuous and strictly increasing CDF F can be reduced to the uniform case by a change of variables.

³We drop the argument s for allocation/audit rules that are deterministic for almost every $\mathbf{t}$.

⁴This can be shown using arguments similar to those in Ben-Porath et al. (2014) and Erlanson and Kleiner (2020).

henceforth we will do that.⁵ It will also be useful to formulate this mechanism design problem in terms of interim rules. Given a mechanism (p, a) , its *interim rules* are defined for all $t \in T$ by

$$\begin{aligned} P(t) &:= \mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t, \mathbf{t}_{-i}, s)] \\ A(t) &:= \mathbb{E}_{\mathbf{t}_{-i}, s}[a_i(t, \mathbf{t}_{-i}, s)]. \end{aligned}$$

We say that p *induces* P and a *induces* A .

We can then simplify the Bayesian incentive compatibility constraints. First, since $u_i(t') > 0$ for all $i \in N$ and all $t' \in T$, $u_i(t')$ cancels from both sides of equation (1) above. Thus, a mechanism is BIC if and only if, for all $t, t' \in T$,

$$\mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t', \mathbf{t}_{-i}, s)] \geq \mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t, \mathbf{t}_{-i}, s)(1 - a_i(t, \mathbf{t}_{-i}, s))] \quad (2)$$

Note that, to ensure that agent i reports truthfully, it is only helpful to audit agent i if $p_i(t, \mathbf{t}_{-i}, s) = 1$. Otherwise, the audit has no impact on the expected probability of being allocated an object when reporting a lie. Using this insight, we can further simplify the incentive constraints expressed above in (2) by correlating the allocation and audit rule. After the random variable s is drawn, either agent i is *scheduled* to get an object $p_i(\mathbf{t}, s) = 1$ or not $p_i(\mathbf{t}, s) = 0$, since we cannot penalize an agent that is not scheduled to get an object we can restrict attention to only audit agents that are scheduled to get an object $p_i(\mathbf{t}, s) = 1$. Therefore, without loss of generality we can assume that $a_i(\mathbf{t}, s) = 1$ only if agent i is scheduled to get an object $p_i(\mathbf{t}, s) = 1$. Henceforth, we restrict attention to mechanisms satisfying this. Using this observation, $\mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t, \mathbf{t}_{-i}, s)a_i(t, \mathbf{t}_{-i}, s)] = \mathbb{E}_{\mathbf{t}_{-i}, s}[a_i(t, \mathbf{t}_{-i}, s)]$ and the right-hand side of (2) simplifies to $P(t) - A(t)$.

These observations imply the following characterization of Bayesian incentive compatibility.

Lemma 1. *A symmetric mechanism (p, a) is BIC if and only if its interim rules (P, A) satisfy, for all $t \in T$,*

$$A(t) \geq P(t) - \inf_{t' \in T} P(t'). \quad (3)$$

3 Optimal mechanisms

In this section, we define the class of merit-with-guarantee mechanisms and show that an optimal mechanism can always be found in this class.

⁵That is, we focus in the following on mechanisms satisfying, for all $\mathbf{t}$ and permutations σ , $\mathbb{E}_s[p_i(\mathbf{t}, s)] = \mathbb{E}_s[p_{\sigma(i)}(\mathbf{t}_\sigma, s)]$, and similarly for a and pa .

3.1 Merit-with-guarantee mechanisms

We begin by discussing some benchmark allocation rules. The first-best rule allocates objects to the agents with the m highest types. However, if only $k < m$ agents can be verified, this rule cannot be part of an incentive-compatible mechanism: an agent with the lowest type would receive an object with probability zero by reporting truthfully, but with positive probability by reporting a higher type because not every agent receiving an object can be verified.

A simple incentive-compatible alternative is to allocate objects only to the k highest reports, verifying these reports, and allocating the objects if the reports are truthful. The remaining $m - k$ objects are then allocated randomly among the remaining agents. Incentive constraints are slack in this mechanism: any misreport can affect the allocation only if it is among the k highest reports, in which case it is verified with probability one and yields zero probability of receiving an object, whereas truthful reporting yields a strictly positive probability of receiving an object.

Since incentive constraints are slack, the principal can improve upon this mechanism by sometimes allocating more than k objects efficiently while verifying only some of those reports. One way to do so is to introduce a cutoff and allocate objects to the agents with the m highest reports whenever these reports exceed the cutoff, verifying k of these reports, and allocating any remaining objects randomly below the cutoff. This mechanism is incentive compatible for sufficiently high cutoffs. The optimal mechanism shares this feature, but introduces an intermediate region in which only the k highest reports obtain an object.

We now define the class of merit-with-guarantee mechanisms.

Definition 2. A mechanism (p, a) is a merit-with-guarantee mechanism if there exist thresholds $0 < x_1 \leq x_2 < x_3 < 1$ that partition the type space into three regions: $T^\ell := [0, x_1]$, $T^k := [x_2, x_3]$, and $T^m := (x_1, x_2) \cup (x_3, 1]$, such that the allocation rule p can be decomposed as $p = p^M + p^\ell$, where:

- (i) $p_i^M(\mathbf{t}) = 1$ if either (a) $t_i \in T^m$ and t_i is among the m highest reports in $\mathbf{t}$, or (b) $t_i \in T^k$ and t_i is among the k highest reports in $\mathbf{t}$; otherwise $p_i^M(\mathbf{t}) = 0$,⁶
- (ii) p^ℓ assigns zero probability to agents with reports in T^m : $p_i^\ell(t_i, \mathbf{t}_{-i}) = 0$ whenever $t_i \in T^m$; moreover, for some constant $\varphi \in (0, \frac{m}{n})$, $\mathbb{E}_{\mathbf{t}_{-i}}[p_i^\ell(t_i, \mathbf{t}_{-i})] = \varphi$ for all i and all $t_i \in T^\ell \cup T^k$; and
- (iii) a randomly selects up to k agents so that $\mathbb{E}_{\mathbf{t}_{-i}}[a_i(t_i, \mathbf{t}_{-i})] \geq \mathbb{E}_{\mathbf{t}_{-i}}[p_i(t_i, \mathbf{t}_{-i})] - \varphi$ for all i and all $t_i \in T$.

⁶If $t_i = t_j$ for some i, j , we use an arbitrary tie-breaking rule.

Any merit-with-guarantee mechanism can be implemented in two stages: In the first stage, objects are scheduled to be allocated according to the *merit-based allocation rule* p^M : those agents whose reports are among the m highest reports and lie in T^m and those whose reports are among the k highest reports and lie in T^k are scheduled to obtain an object; k of those agents are randomly chosen by the audit rule for a verification; if their reports are truthful (which they will be in equilibrium), the agents that are scheduled for allocation obtain their object (in case of a lie being detected, the lying agent does not receive an object). Objects not allocated in this first stage enter the second-stage lottery. All agents with reports in T^ℓ , as well as agents with reports in T^k who are not scheduled for allocation in the first stage, participate in this lottery. In this second stage, the *lottery-based allocation rule* p^ℓ allocates all remaining objects so that all agents with types in T^ℓ and T^k have the same expected probability of receiving an object in the lottery stage.

To satisfy property (iii), it is sufficient for a merit-with-guarantee mechanism to only audit agents who are scheduled for allocation in the first stage, i.e., agents for whom $p_i^M(\mathbf{t}) = 1$. In that case, low reports that lie in T^ℓ will never be audited. Moreover, all agents with reports in T^k who are scheduled to receive an object in the first stage will be audited for sure: since $\mathbb{E}_{\mathbf{t}_{-i}}[a_i(t_i, \mathbf{t}_{-i})] \geq \mathbb{E}_{\mathbf{t}_{-i}}[p_i^M(t_i, \mathbf{t}_{-i})]$ and only agents scheduled for first-stage allocation are audited, we obtain $a_i(\mathbf{t}) = p_i^M(\mathbf{t})$ for $t_i \in T^k$. On the other hand, because up to m objects are scheduled for allocation to types in T^m and at most k of the reports can be verified, the merit-with-guarantee mechanism has to carefully balance incentives for low types to be truthful. We describe incentives in merit-with-guarantee mechanisms in more detail below.

Figure 1 illustrates a merit-with-guarantee mechanism. In panel (a), $x_1 = x_2$ and therefore T^m is an interval; in panel (b), $x_1 < x_2$ and therefore T^m is the union of two intervals. When T^m consists of a single interval, the principal allocates objects efficiently within the T^m region. Only if fewer than m agents have reports in T^m does the principal allocate objects to agents with reports in T^k who are among the k highest reports. Any remaining objects enter the lottery stage and are allocated among agents with reports in T^ℓ and T^k who did not obtain an object in the first stage. Example 1 illustrates how the merit-with-guarantee mechanism allocates objects and determines audits.

Example 1. Suppose that three objects and two audits are available ($m = 3$, $k = 2$).

If at least three reports are in T^m , the audit rule randomly selects two of the three highest reports for verification. If no lie is detected, all three objects are allocated to the agents with the three highest reports, and no object remains for the second-stage lottery.

Suppose instead that exactly one report is in T^m and at least one report is in T^k . Then the report in T^m and the highest report in T^k are verified, and—if no lie is detected—both agents receive an object. The remaining object is allocated in the second stage via a lottery

among all unassigned agents with reports in T^ℓ and T^k .

When T^m consists of two intervals, objects may already be allocated inefficiently in the first stage. In particular, an agent with a report in the lower T^m region (i.e., in (x_1, x_2)) who is among the m highest reports receives an object, while an agent with a report in T^k who is among the m highest but not among the k highest reports does not. As before, any objects that remain unallocated after the first stage enter the lottery and are allocated among agents with reports in T^ℓ and T^k who did not obtain an object in the first stage.

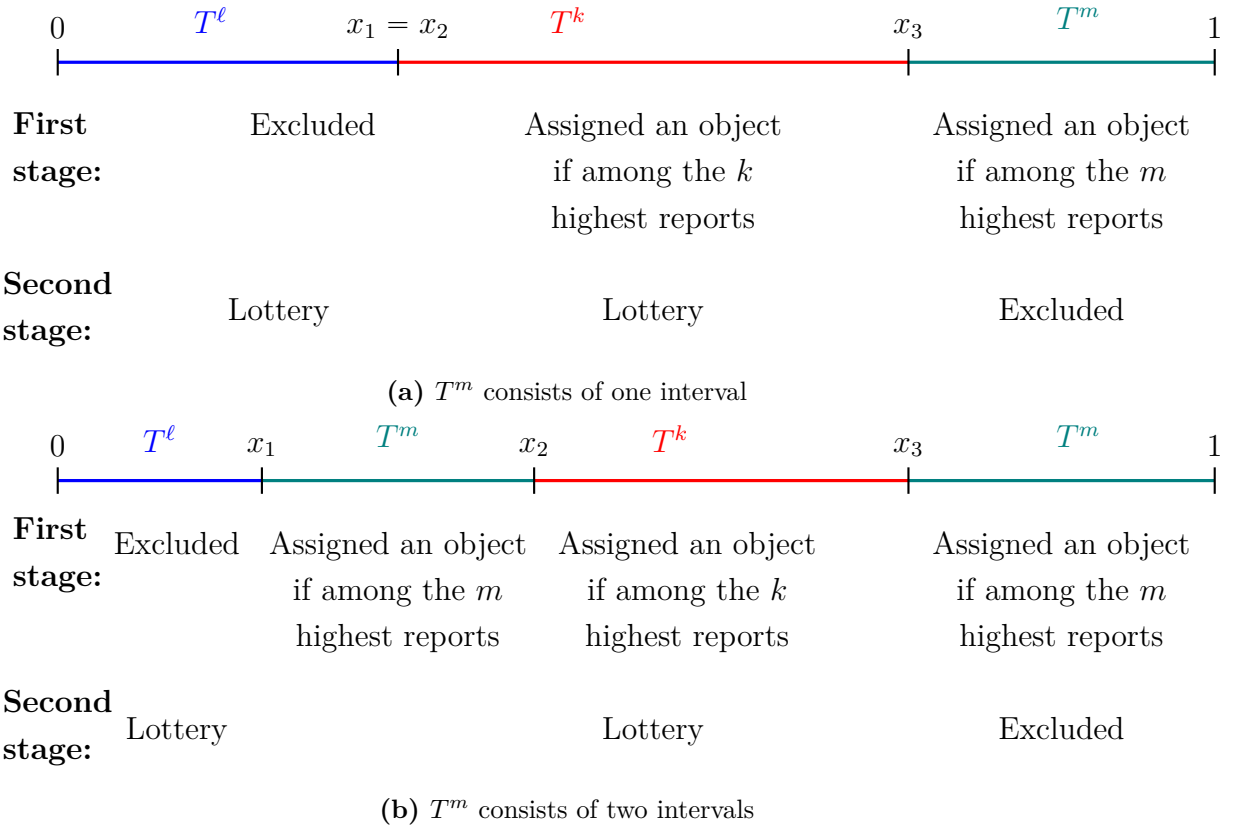


Figure 1: Allocation rule in a merit-with-guarantee mechanism. Panel (a) illustrates a case where T^m is an interval, panel (b) a case where T^m is the union of two intervals.

3.2 A merit-with-guarantee mechanism is optimal

The following result shows that a merit-with-guarantee mechanism is optimal.

Theorem 1. *A merit-with-guarantee mechanism is optimal. Moreover, every optimal and symmetric mechanism equals a merit-with-guarantee mechanism almost surely.*

A key feature of a merit-with-guarantee mechanism is that even low types obtain an object with some probability $\varphi > 0$, which we call the *guarantee* of the mechanism. To

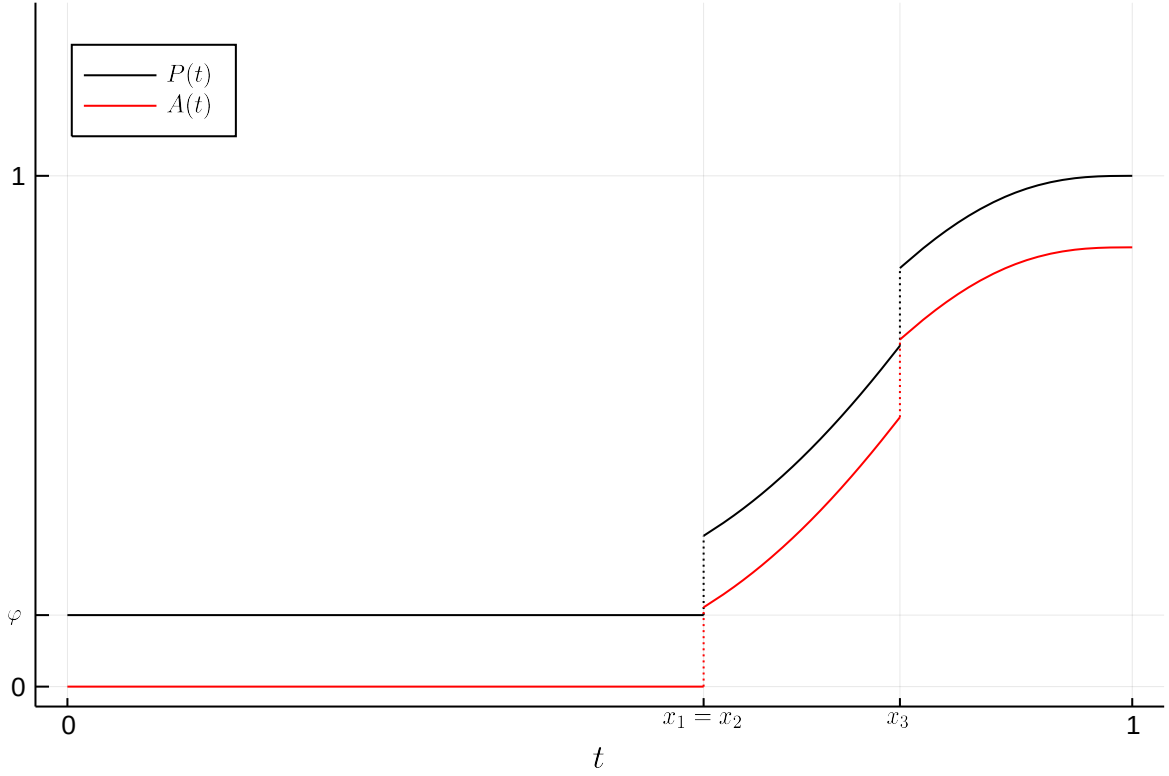


Figure 2: This figure illustrates the induced interim allocation and audit rules from the optimal merit-with-guarantee mechanism in an example where $n = 8, m = 3, k = 2$ and $\varphi = 0.14$

understand incentives for truthful reporting, consider an agent with a type $t_i \in T^\ell$. Under truthful reporting, this agent obtains an object only in the lottery stage, which happens with probability φ .

If the agent misreports as a type in T^k , he is scheduled to receive an object in the first stage only if his report is among the k highest. In that case, the report is audited with probability one, and the agent receives no object because the lie will be detected. Since the lottery probability is equal to φ for all reports in T^ℓ and T^k , such a deviation is not profitable.

If the agent instead misreports as a type in T^m , he may obtain an object in the first stage if more than k agents report types in T^m , since not all such reports can be verified and the lie might not be detected. However, agents reporting in T^m do not enter the lottery stage, so the agent gives up on the probability of obtaining an object in the lottery stage. Provided that T^m is not too large, the probability that more than k agents report in T^m is small. Therefore, an agent who sends a misreport that is in T^m faces a high risk that his lie is detected; taking expectations over the other agents' reports, it is therefore optimal for type t_i to report truthfully.

While it is optimal, in expectation over the others' reports, for each agent to report

truthfully, this is not necessarily optimal if an agent knew the others' reports. To see this, consider an agent i and a profile $\mathbf{t}$ such that m other agents have types in T^m strictly above t_i . If all agents are truthful, the m highest reports receive an object; since agent i 's report is not among the m highest ones, he receives no object. If, instead, he misreports as a type in T^m that is among the m highest reports, he receives an object whenever he is not audited. Since not all m agents can be audited, agent i increases his probability of receiving an object at the type profile $\mathbf{t}$ and this is a profitable deviation. Thus, the merit-with-guarantee mechanism is generally not ex-post incentive compatible. In fact, the proof of Theorem 1 implies that this is not just a feature of the specific merit-with-guarantee mechanism, but that the optimal mechanism cannot satisfy these stronger incentive constraints.

Remark 1. *The optimal mechanism cannot be implemented to satisfy ex-post incentive constraints.*

To gain intuition for why a merit-with-guarantee mechanism achieves a high payoff for the principal, recall that allocating objects only to the k highest types and distributing the remaining $m - k$ objects randomly leaves incentive constraints slack. A merit-with-guarantee mechanism improves upon this by sometimes allocating more than k objects to high reports—specifically, by allocating up to m objects when reports are in T^m . Incentive compatibility is preserved as long as the probability that more than k agents report types in T^m is sufficiently small.

In the optimal merit-with-guarantee mechanism, the region T^m cannot be expanded further without violating incentive constraints for low types. At the same time, when fewer than k objects are allocated to agents in T^m , some verification capacity remains unused. The optimal mechanism addresses this by allocating some of the remaining objects to agents with reports in T^k and auditing these reports with probability one. While this intuition for optimality is incomplete, it captures insights into why such mechanisms are optimal. Taking the perspective of interim rules in Section 4 will provide deeper insights into why this mechanism is optimal and why the regions take the specific forms: T^ℓ and T^k are intervals, and T^m is the union of at most two intervals.

If T^m is the union of two disjoint intervals, then even the merit-based allocation rule can allocate objects inefficiently: an agent with report in T^k might not get an object in the first stage even though an agent with a lower report will get an object in the first stage. Economic intuition suggests that such a mechanism might be improved by shifting allocations towards higher types by enlarging the higher interval in T^m and shrinking the lower interval in T^m . However, misreports in the higher interval in T^m are particularly attractive for low types as they result in high allocation probabilities, and the mechanism needs sufficiently many audits to deter such misreports. In general, we are not able to

show that sufficiently many audits are available to turn such a perturbed allocation rule into an incentive-compatible mechanism.

Nevertheless, we are able to show that in many cases the optimal merit-with-guarantee mechanism uses a partition where T^m is an interval, so that the mechanism simplifies to one with two cutoffs determining the merit-based allocation rule. Proposition 3 in the Appendix provides sufficient conditions for this. These conditions are easy to check, and depend only on the number of agents, objects, and audits available; they do not depend on the valuation function v . To illustrate that these conditions are useful, we verified computationally that the conditions are satisfied whenever $n \leq 100$, which yields the following result:

Corollary 1. *In the optimal merit-with-guarantee mechanism, T^m is an interval whenever $n \leq 100$.*

Theorem 1 shows that an optimal mechanism can always be found in the class of merit-with-guarantee mechanisms, and that (at least among symmetric mechanisms) only merit-with-guarantee mechanisms can be optimal.⁷ Note that the definition of a merit-with-guarantee mechanism does not fully specify the allocation and audit rules; for the lottery-based allocation, it only specifies that the probability of obtaining an object in the lottery stage is constant across types in $T^\ell \cup T^k$. One reason why we leave the exact implementation unspecified is that any lottery-based allocation that induces constant interim rules is optimal. If we want to discuss whether a given mechanism is optimal, this is the only implication we can draw from our analysis about the lottery-based allocation. In this sense, Theorem 1 captures all the qualitative insights we can gain into optimal mechanisms.

Another reason why we leave the exact implementation unspecified is that typically there are many specifications and there does not appear to be a canonical one. A natural approach would be to allocate all objects available in the lottery stage uniformly at random among the agents entering the lottery. While this often yields winning probabilities that are almost constant, this does not work precisely in general. The reason is that agents with reports in T^k have a higher probability of receiving an object in the merit stage, their probability of entering the lottery is decreasing in their report. On the other hand, the expected number of objects entering the lottery stage conditional on an agent with type $t \in T^k$ entering the lottery is also not constant in t . Therefore, to maintain the same ex-ante probability φ , the conditional probability of winning the lottery cannot be uniform across agents. Our proof establishes that one can always implement the lottery in a way that yields constant winning probabilities of φ .⁸

⁷In fact, one can show that even among asymmetric mechanisms, essentially only merit-with-guarantee mechanisms can be optimal.

⁸One way to construct optimal ex post rules p and a follows from observing that the interim rules

4 Why a merit-with-guarantee mechanism is optimal

In this section, we explain in more detail why a merit-with-guarantee mechanism is optimal. Along the way, we will obtain additional insights into the structure of optimal mechanisms and show how an optimal merit-with-guarantee mechanism can be found by optimizing over a single parameter. To do all this, it will be useful to formulate the problem in terms of interim rules because this significantly reduces its dimensionality. We will see that the resource constraints imply that any interim rules must satisfy specific conditions, which in turn determine the shape of the various allocation regions in a merit-with-guarantee mechanism. We formulate an upper bound on the principal's payoff and show that a particular merit-with-guarantee mechanism might attain this bound.

In the following, we define which audit rules are feasible. Recall our convention that in any mechanism $a_i(\mathbf{t}, s) = 1$ only if $p_i(\mathbf{t}, s) = 1$, which is without loss of generality and which we used to simplify incentive constraints.

Definition 3. *An audit rule a is feasible given an allocation rule p if $a_i(\mathbf{t}, s) \leq p_i(\mathbf{t}, s)$ for all $i \in N$, $\mathbf{t} \in \mathbf{T}$, and $s \in [0, 1]$. A pair of interim rules (P, A) is feasible if there exist an allocation rule p and an audit rule a which is feasible given p that induce (P, A) .*

We can now reformulate the principal's problem using interim rules as follows:

$$\begin{aligned} \max_{P, A} \quad & n \mathbb{E}_t[P(t)v(t)] \\ \text{s.t.} \quad & (P, A) \text{ is feasible and} \\ & A(t) \geq P(t) - \inf_{t' \in T} P(t') \text{ for all } t \in T. \end{aligned} \tag{Opt}$$

To find the optimal interim rules, we start by deriving constraints that any feasible mechanism must satisfy. These constraints are due to the limited availability of objects and audits (for related conditions, see Border 1991).

A feasibility constraint on any interim allocation rule P , which stems from at most m objects being available, is that for any type t ,

$$n \int_t^1 P(x) dx \leq c^{allo}(t) \tag{4}$$

where

$$c^{allo}(t) := \sum_{i=1}^n \min\{i, m\} \binom{n}{i} (1-t)^i t^{n-i}.$$

P and A are mixtures of the extreme points characterized in Kleiner, Moldovanu and Strack (2021). Following the construction in the proof of their Theorem 3, one obtains particular ex post rules that are mixtures of ex post allocation rules that contain pooling intervals (an analogous argument is used by Maskin and Riley (1984) for the single-good case to prove their Theorem 7). We illustrate another explicit construction in the context of Example 2.

To see this, note that the left-hand side of (4) denotes the expected number of agents with a type above t that are assigned an object. For any $i \in \{1, \dots, n\}$, the probability that there are i agents with a type above t is $\binom{n}{i}(1-t)^i t^{n-i}$, and at most $\min\{i, m\}$ of these agents can be assigned an object. By summing over i , we obtain (on the right-hand side) an upper bound on the expected number of agents with a type above t that can be assigned an object, and the inequality follows.

We have seen in Lemma 1 that whether a mechanism is BIC depends in particular on the lowest probability with which some type obtains an object; for an interim allocation rule P , let therefore $\varphi := \inf_{t'} P(t')$. We then obtain that for any type t ,

$$n \int_t^1 P(x) dx \leq c_\varphi^{ic}(t), \quad (5)$$

where

$$c_\varphi^{ic}(t) := m - nt\varphi.$$

Similarly to inequality (4), the left-hand side in (5) denotes the expected number of agents with a type above t receiving an object. Because there are m objects and because in expectation nt agents have a type below t and each of them receives an object with a probability of at least φ , the inequality follows.⁹

Since at most k audits are available, we obtain in analogy with inequality (4) the following feasibility constraint on A that must hold for all types t :

$$n \int_t^1 A(x) dx \leq \sum_{i=1}^n \min\{i, k\} \binom{n}{i} (1-t)^i t^{n-i}.$$

Lemma 1 implies that if (P, A) is BIC then $A(t) \geq P(t) - \inf_{t'} P(t')$. Plugging this into the inequality above and using $\varphi = \inf_{t'} P(t')$, we obtain that for any type t ,

$$n \int_t^1 P(x) dx \leq c_\varphi^{aud}(t), \quad (6)$$

where

$$c_\varphi^{aud}(t) := \sum_{i=1}^n \min\{i, k\} \binom{n}{i} (1-t)^i t^{n-i} + n(1-t)\varphi.$$

Finally, we combine the constraints in inequalities (4), (5), and (6) to obtain

$$n \int_t^1 P(x) dx \leq c_\varphi(t) := \min\{c^{allo}(t), c_\varphi^{aud}(t), c_\varphi^{ic}(t)\}. \quad (7)$$

One can view inequality (7) as being parametrized by φ ; any feasible and BIC pair of interim rules (P, A) with $\inf_{t'} P(t') = \varphi$ must satisfy this parametrized inequality.

⁹Formally, the inequality follows because (4) implies $n \int_0^t P(x) dx + n \int_t^1 P(x) dx \leq m$ and $\varphi \leq P(t)$ by definition.

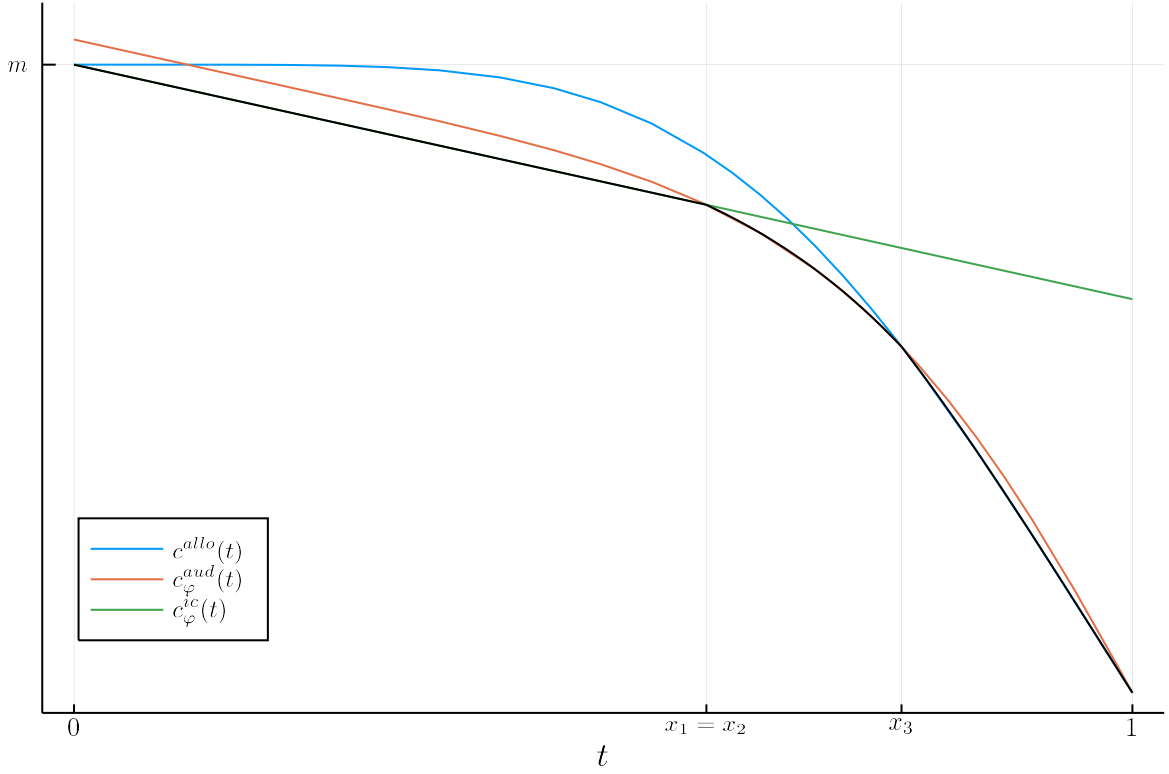


Figure 3: This figure illustrates the right-hand sides of (4)–(6) as functions of t for an example where $n = 8$, $m = 3$, $k = 2$ and $\varphi = 0.14$.

Figure 3 shows the right-hand sides of (4)–(6) for an example where $n = 8$, $m = 3$, $k = 2$ and $\varphi = 0.14$. As can be seen in the figure, for an interval of high types (for $t \geq x_3$) constraint (4) is the tightest, for an interval of intermediate types (for $t \in [x_1, x_3]$) constraint (6) is the tightest, and for an interval of low types (for $t < x_1$) constraint (5) is the tightest. In fact, these regions of tightest constraints always take either this form, or have one additional interval on which (4) is the tightest constraint, as is shown in Lemma 7 in Appendix A.3.

These constraints restrict the principal in how often she can allocate to agents with a type above any t . Since the principal aims to allocate as often as possible to high types, allocation rules that satisfy the tightest constraint with equality for every t are particularly attractive, as formalized in the following result and illustrated in Figure 4.

Lemma 2. *For any feasible mechanism $(\tilde{p}, \tilde{a})$ satisfying $\mathbb{E}_{t-i}[\tilde{p}_i(t, \mathbf{t}_{-i})] \geq \varphi$, the principal's payoff is at most*

$$\int_0^1 c_\varphi(t) dv(t).$$

It achieves this upper bound if and only if (7) holds as equality for all t .

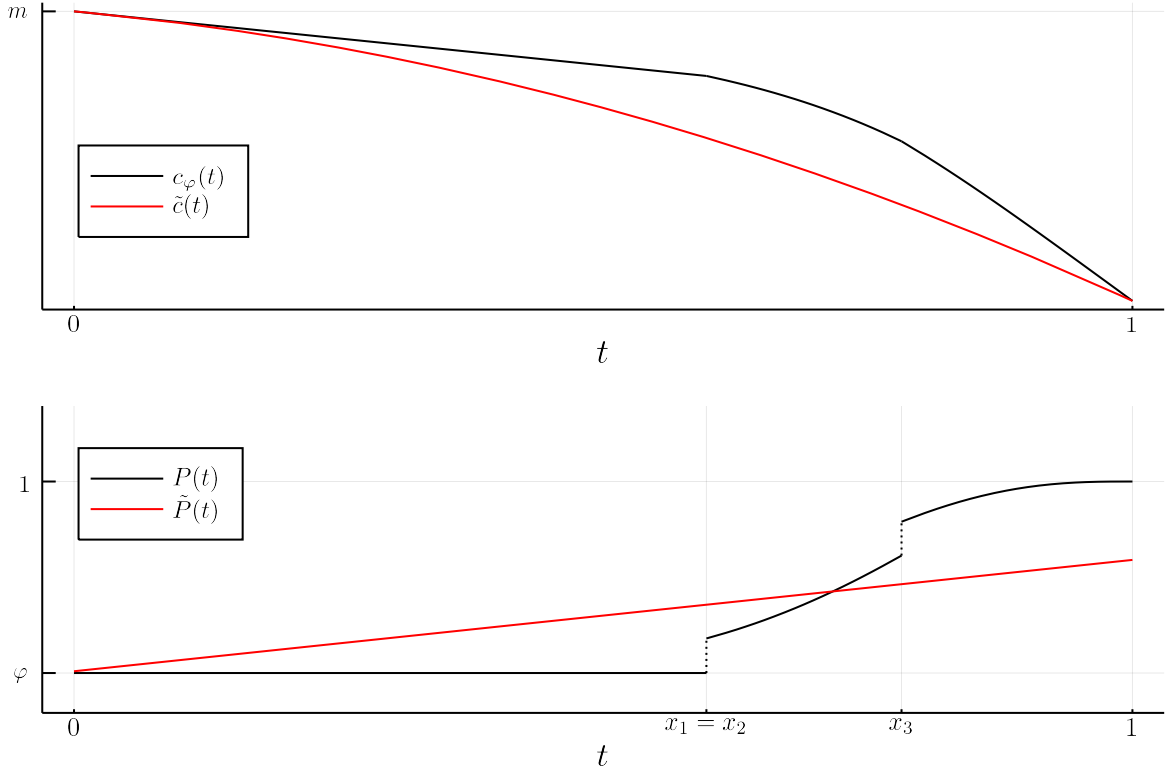


Figure 4: The upper panel illustrates the left-hand and right-hand side of (7) for an arbitrary mechanism $\tilde{P}$. The lower panel shows the interim allocation rule $\tilde{P}$ and the hypothetical allocation rule that makes (7) binding, illustrating how $\tilde{P}$ allocates less often to high types.

Proof. Using integration by parts, we obtain

$$\begin{aligned} n \int_0^1 \tilde{P}(t) v(t) dt &= n v(t) \int_0^t \tilde{P}(x) dx \Big|_{t=0}^1 - n \int_0^1 \int_0^t \tilde{P}(x) dx dv(t) \\ &= n \int_0^1 \int_t^1 \tilde{P}(x) dx dv(t) \leq \int_0^1 c_\varphi(t) dv(t) \end{aligned}$$

where the last inequality follows from (7). Equality holds if and only if (7) is an equality for all t . $\square$

We now illustrate in the context of the example in Figure 3 why a merit-with-guarantee mechanism induces (7) to be binding everywhere (and why only merit-with-guarantee mechanisms can do so). So consider a merit-with-guarantee mechanism (p, a) with $T^\ell = [0, x_1]$, $T^k = (x_1, x_3]$ and $T^m = [x_3, 1]$, and suppose it satisfies $\mathbb{E}_{t-i}[p_i(t, \mathbf{t}_{-i})] \geq \varphi$ for all $t \in T$. We verify that this mechanism satisfies (7) as an equality for every t : For any $t \geq x_3$, under this allocation rule the expected number of objects allocated to agents with types above t is $\sum_{i=1}^n \min\{i, m\} \binom{n}{i} (1-t)^i t^{n-i}$, which equals $c^{allo}(t)$; hence, (7) is an equality for all $t \in T^m$. Also, for any $t \in T^k$, the expected number of objects allocated to

agents with types above t is

$$\sum_{i=1}^n \min(i, k) \binom{n}{i} (1-t)^i t^{n-i} + \sum_{i=1}^n [\min(i, m) - \min(i, k)] \binom{n}{i} (1-x_3)^i x_3^{n-i} + n\varphi(x_3 - t);$$

the first sum captures that the k highest agents get an object if their type is above t , the second sum captures that up to m objects are allocated to types above t if their types lie in T^m , and the last term captures the expected allocation to agents with types above t in the lottery stage. Using the definitions of c^{allo} and c_φ^{aud} , we can simplify the last display to

$$c_\varphi^{aud}(t) + c^{allo}(x_3) - c_\varphi^{aud}(x_3) = c_\varphi^{aud}(t),$$

where the last equality uses that, by definition, $c^{allo}(x_3) = c_\varphi^{aud}(x_3)$. Hence, for any $t \in [x_1, x_3)$, (7) holds as an equality. Finally, using that $c_\varphi^{aud}(x_1) = c_\varphi^{ic}(x_1)$ by definition of x_1 , one can verify that (5) is binding for $t \in [0, x_1]$. Jointly, these observations imply that if the merit-with-guarantee mechanism illustrated in Figure 3 satisfies $\mathbb{E}_{t-i}[p_i(t, \mathbf{t}_{-i})] \geq \varphi$ for all $t \in T$, then it is optimal; moreover, any optimal mechanism must take this form almost everywhere.

For the general proof, we consider for a given φ the sets T^m , T^ℓ , and T^k , defined as above as the set of types for which (4), (5), and (6) are tightest, respectively. We then define the merit-based allocation rule p^M based on these sets: an agent gets an object if either her type is among the m highest ones and her report is in T^m , or her report is in T^k and among the k highest ones. We show in Lemma 9 that we can complement this allocation rule by a lottery-based rule for the second stage so that each type has a probability at least φ of getting an object under the combined rule p ; moreover, (7) will be binding for all t . Lemma 12 implies that there exists an audit rule a that is feasible given p such that (p, a) is incentive compatible. Lemma 2 then implies that (p, a) is optimal.

5 Optimal guarantee and large market analysis

5.1 Optimal guarantee

In this section, we discuss how to optimally choose the cutoffs x_1, x_2, x_3 for the merit-with-guarantee mechanism. The proof of Theorem 1 reveals that finding the optimal cutoffs can be reduced to optimizing the guarantee φ , a single parameter, and then using the cutoffs that are induced by φ as follows: Given any $\varphi \in [0, \frac{m}{n}]$, we define the cutoffs $x_1(\varphi), x_2(\varphi)$ and $x_3(\varphi)$ to satisfy $c_\varphi^{ic}(t) \leq \min\{c_\varphi^{aud}(t), c^{allo}(t)\}$ for all $t \in (0, x_1(\varphi))$, $c_\varphi^{aud}(t) \leq \min\{c_\varphi^{ic}(t), c^{allo}(t)\}$ for all $t \in (x_2(\varphi), x_3(\varphi))$, and $c^{allo}(t) \leq \min\{c_\varphi^{ic}(t), c_\varphi^{aud}(t)\}$ for all $t \in (x_1(\varphi), x_2(\varphi)) \cup (x_3(\varphi), 1]$. The existence of these cutoffs follows from Lemma 7.

Which guarantees φ can be optimal? It turns out that for optimal values of φ , all three regions are nonempty and we can describe optimal values using a simple first-order condition.

Proposition 1. *In the optimal merit-with-guarantee mechanism, T^m , T^k , and T^ℓ are nonempty: $0 < x_1 \leq x_2 < x_3 < 1$. Moreover, every optimal choice of φ must satisfy the following first-order condition:*

$$x_1(\varphi)v(x_1(\varphi)) + (1 - x_2(\varphi))v(x_2(\varphi)) - (1 - x_3(\varphi))v(x_3(\varphi)) = \int_0^{x_1(\varphi)} v(t) dt + \int_{x_2(\varphi)}^{x_3(\varphi)} v(t) dt. \quad (8)$$

The first-order condition balances two effects of a marginal increase in φ . The right-hand side captures the direct benefit: a higher guarantee increases allocation probabilities via the lottery rule for all types in T^ℓ and T^k . The left-hand side captures the net effect on the principal's payoff from changing the partition $\{T^\ell, T^k, T^m\}$ through the movement of the boundaries x_1 , x_2 , and x_3 . The first term, $x_1v(x_1)$, reflects that as φ increases, the boundary x_1 shifts to the right: types near x_1 move from merit-based allocation into the lottery region, receiving the object with probability φ . The second term, $(1 - x_2)v(x_2)$, reflects that the lower boundary of T^k shifts to the right, shrinking the audit region from below. The third term, $-(1 - x_3)v(x_3)$, enters negatively because the upper boundary of T^k shifts to the left, expanding the upper interval of T^m , where objects are allocated efficiently. At the optimal value of φ , where all three regions are nonempty, increasing φ expands both T^ℓ and T^m at the expense of T^k . The first-order condition equates the direct benefit with the net boundary effect from this reallocation of types across regions.

We briefly explain why all three regions are used in an optimal mechanism. To see that T^m is nonempty, note that because there are only m objects, φ can be at most $\frac{m}{n}$. Moreover, $\varphi = \frac{m}{n}$ corresponds to allocating uniformly at random, which is dominated by slightly smaller values of φ . Indeed, whenever $\varphi < \frac{m}{n}$, c_φ^{allo} is strictly below c_φ^{ic} and c_φ^{aud} in a neighborhood around $t = 1$, and T^m is nonempty and allocations are efficient for the highest types.

To see that T^ℓ is nonempty, suppose to the contrary that it is empty. This implies $c_\varphi^{ic}(0) \geq c_\varphi^{aud}(0)$. A slightly higher value of φ increases c_φ^{aud} pointwise, which expands T^m (see Figure 3) and increases allocation probabilities for higher types at the cost of types close to 0, improving the principal's payoff and contradicting optimality.

Finally, to see that T^k is nonempty, suppose φ is high enough that c_φ^{aud} lies everywhere above either c_φ^{ic} or c_φ^{allo} , so that T^k is empty. Then reducing φ lowers c_φ^{aud} pointwise, expanding T^m , since the intersection between c_φ^{ic} and c_φ^{allo} shifts to the left, and increasing allocation probabilities for higher types at the cost of lower types. As φ decreases further, T^k eventually consists of a single point. Beyond this, reducing φ introduces a tradeoff: T^m shrinks, so efficient allocation occurs for fewer types, but T^ℓ also shrinks, so allocation

becomes more efficient for intermediate types. The proof of Proposition 1 shows via the first-order condition that small reductions in φ remain beneficial, implying that T^k is nondegenerate at the optimum.

Example 2. Suppose that there are three agents, two objects to allocate, and only one agent can be verified: $n = 3, m = 2, k = 1$. We assume $v(t) = t$ for all t .

We now illustrate how to use Proposition 1 to find an optimal value of φ . This corresponds to determining which constraint function is binding for the upper bound,

$$3 \int_t^1 P(x) dx \leq c_\varphi(t) = \min\{c^{allo}(t), c_\varphi^{aud}(t), c_\varphi^{ic}(t)\}.$$

For this example the constraint functions are the following:

$$c^{allo}(t) = t^3 - 3t^2 + 2, \quad c_\varphi^{aud}(t) = -t^3 - 3\varphi t + 1 + 3\varphi, \quad \text{and} \quad c_\varphi^{ic}(t) = 2 - 3\varphi t,$$

where the optimal value of φ lies in $[\frac{1}{3}, \frac{2}{3}]$. By Corollary 1, $T^m = [x_3, 1]$, giving us the following structure of the merit-with-guarantee rule:

$$P(t) = \begin{cases} \varphi & \text{for } t \in [0, x_1(\varphi)) \\ t^2 + \varphi & \text{for } t \in [x_1(\varphi), x_3(\varphi)) \\ 2t - t^2 & \text{for } t \in [x_3(\varphi), 1]. \end{cases}$$

The expected per-agent payoff for the principal as a function of φ is

$$U(\varphi) = \int_0^{x_1(\varphi)} \varphi t dt + \int_{x_1(\varphi)}^{x_3(\varphi)} t^3 + \varphi t dt + \int_{x_3(\varphi)}^1 2t^2 - t^3 dt.$$

Using Proposition 1 we have the following condition for φ to be optimal:

$$x_1(\varphi) = x_3(\varphi)[1 - x_3(\varphi)] + \int_0^{x_3(\varphi)} t dt.$$

Solving this equation gives us the optimal value $\varphi^* \approx 0.348$ and an expected payoff of 1.223. This can be compared with the expected payoff of 1 from using a random allocation rule where all agents get an object with an equal probability of $\frac{2}{3}$. Another comparison is with the first-best mechanism that always allocates the two objects to the agents with the highest types, yielding an expected payoff of 1.25.

We now illustrate one construction of a lottery rule p^ℓ for this example. At the lottery stage, all agents with reports in T^ℓ are treated symmetrically: their winning probability depends only on the number of lottery participants, the number of available objects, and whether any participant reports a type in T^k . For lottery instances involving only agents from T^ℓ , if two agents compete for one object, each wins with probability $1/2$; if three such agents compete for two objects, each wins with probability $2/3$. When one agent reports

from T^ℓ and another from T^k , with one object available, we set the winning probabilities to 0.6872 for the T^k agent and 0.3128 for the T^ℓ agent. This ensures that all types in T^ℓ receive an object in the lottery stage with ex-ante probability φ^* .

To complete the description of the lottery rule, we must specify winning probabilities when both agents report types in T^k . Since agents in T^k enter the lottery only when they fail to receive an object in the merit stage, the probability of entering the lottery decreases with their report (higher reports are more likely to rank among the top k). Therefore, to ensure all reports in T^k obtain an object at the lottery stage with ex-ante probability φ^* , the conditional winning probability in the lottery must increase in their report. When two agents with reports $t, t' \in T^k$ both enter the lottery, we set the winning probability for the agent with report t to $r(t, t') = \frac{1}{2} + g(t) - g(t')$ for some strictly increasing function g . This ensures that the object is always assigned: $r(t, t') + r(t', t) = 1$. The function g is uniquely determined by requiring that each agent with report $t \in T^k$ receives an object at the lottery stage with ex-ante probability φ^* .

5.2 Large market analysis

While typical applications we have in mind involve a moderate number of agents and objects, in this section we analyze large markets to better understand limiting properties of the optimal mechanisms. Specifically, consider a sequence of markets indexed by n , where market n has n agents, m_n objects, and audit capacity k_n , with $m_n/n \rightarrow \alpha$ and $k_n/n \rightarrow \beta$ as $n \rightarrow \infty$ for some fixed $0 < \beta < \alpha < 1$. Let P_n denote the interim allocation rule of the optimal merit-with-guarantee mechanism in a market of size n and let φ_n denote the associated optimal guarantee.

Proposition 2. *As $n \rightarrow \infty$, the optimal interim allocation rule satisfies*

$$P_n(t) \rightarrow \begin{cases} \alpha - \beta & \text{if } t < \frac{1-\alpha}{1-\alpha+\beta} \\ 1 & \text{if } t > \frac{1-\alpha}{1-\alpha+\beta} \end{cases}$$

and the optimal interim audit rule satisfies

$$A_n(t) \rightarrow \begin{cases} 0 & \text{if } t < \frac{1-\alpha}{1-\alpha+\beta} \\ 1 - (\alpha - \beta) & \text{if } t > \frac{1-\alpha}{1-\alpha+\beta}. \end{cases}$$

Intuitively, as the market grows, from each agent's perspective there is less uncertainty about whether her type is among the m_n highest ones: the probability is close to zero if $t < 1 - \alpha$ and close to 1 if $t > 1 - \alpha$. As a consequence, the functions $\frac{1}{n}c^{allo}$ converge to a piecewise affine function with a kink at $1 - \alpha$, see Figure 5. For a fixed φ , the same argument applies to c_φ^{aud} with a kink at $1 - \beta$. Since the optimal guarantees φ_n are at least $\frac{m_n - k_n}{n}$, the optimal interim allocation rules converge pointwise to a piecewise

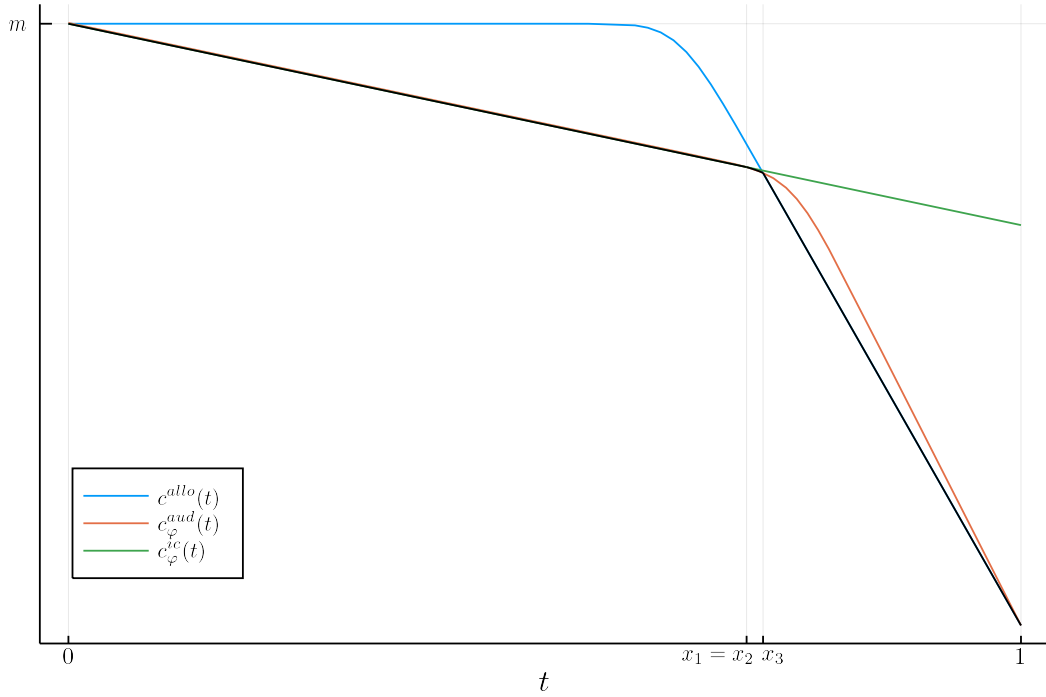


Figure 5: This figure illustrates the right-hand sides of (4)–(6) as functions of t for $n = 250$, $m = 90$, $k = 60$, and $\varphi = 0.1205$. It also shows that in an optimal mechanism for large markets, T^m is an interval and the interval T^k is small.

constant function with a jump at the limit cutoff $\frac{1-\alpha}{1-\alpha+\beta}$, as described in Proposition 2. The fact that incentive constraints are binding for the lowest type implies that the optimal interim audit rule also converges to a piecewise constant function.

Corollary 2. *As α increases, the limit cutoff decreases and the allocation probability for types below this cutoff increases.*

As β increases, the limit cutoff decreases and the allocation probability for types below this cutoff decreases.

As α increases, more objects become available, which allows the principal to decrease the cutoff above which the allocation is efficient. Since this alone would make misreports more attractive, the probability for low reports to receive an object in the lottery increases. As β increases, more audits become available. This allows the principal to reduce the probability for low reports to get an object and thereby to reduce the cutoff for efficient allocation.

Based on Proposition 2, we can describe a simple mechanism for large markets that achieves a per-capita payoff close to that of the optimal mechanism. We reserve $\frac{\beta}{1-\alpha+\beta}n$ objects for allocation in the first round and, uniformly at random, select an equal number of agents from those with reports satisfying $t > \frac{1-\alpha}{1-\alpha+\beta}$ (if fewer agents have such reports, this group is smaller). From this group, we select uniformly at random k_n agents for an

audit and, assuming no lies are detected, allocate an object to all agents in this group. All remaining objects are allocated uniformly at random to agents with reports satisfying $t \leq \frac{1-\alpha}{1-\alpha+\beta}$.

This mechanism is simple in that it uses only a single cutoff and in that it selects agents uniformly at random. As the market size grows, the probability of a type above the cutoff receiving an object converges to 1, and the probability of an audit conditional on being selected for allocation converges to $1 - \alpha + \beta$. So in this mechanism, agents have similar allocation probabilities as in the optimal merit-with-guarantee mechanism in large markets. The merit-with-guarantee rule dominates this simple mechanism in instances when there are many agents with high types because it always allocates efficiently in the highest interval, whereas the simple mechanism rations high types at random and allocates at most $\frac{\beta}{1-\alpha+\beta}n$ objects to agents with high types, and allocates any remaining objects in a lottery among agents with types below the cutoff.

6 Conclusion

We have analyzed the problem of allocating m identical objects to n agents when only k checks are available to verify agents' private information and monetary transfers are unavailable, and have shown that a merit-with-guarantee mechanism is optimal. A key insight is that guaranteeing low types a positive allocation probability φ sustains truthful reporting and in turn allows the mechanism to allocate efficiently beyond the k agents that can be verified.

To illustrate, consider the social housing application discussed in the introduction. A merit-with-guarantee mechanism partitions applicants into three groups based on reported housing need. Those reporting the highest need fall in the region T^m , and are allocated housing in the first stage whenever their reports are among the m highest, even though not all of them can be verified. Applicants reporting moderately high need, in the region T^k , are also allocated housing in the first stage but only when their reports are among the k highest, in which case they are verified with certainty and lose their allocation if found to have misrepresented their circumstances. Applicants reporting lower levels of need, in the region T^ℓ , enter the second-stage lottery together with applicants from T^k whose reports are not among the k highest; all of them receive housing with the guaranteed probability.

Constraints on verification arise naturally in many allocation settings beyond social housing. In the allocation of research grants, for instance, funding bodies rely on peer review to assess proposal quality, but reviewer capacity limits the number of proposals that can be scrutinized in depth. Our results suggest that in such settings, complementing merit-based allocation with a guarantee—a positive allocation probability for agents whose reports are not verified—can improve outcomes by sustaining truthful reporting

while allowing the mechanism to allocate efficiently beyond its verification capacity. While we are not aware of procedures that explicitly advertise such guarantees, they can be interpreted as an implicit feature of systems that combine merit-based assignment with truncated reporting, where applicants below a threshold are pooled and treated symmetrically.

A Proof of Theorem 1

In the remainder of the proof of Theorem 1, we will show that, for any $\varphi \in [0, \frac{m}{n}]$, there exists a merit-with-guarantee mechanism such that the lottery rule satisfies $\mathbb{E}_{\mathbf{t}_{-i}}[p^\ell(t_i, \mathbf{t}_{-i})] = \varphi$ for all $t_i \in T^\ell \cup T^k$, and that there exists an audit rule a so that $\mathbb{E}_{\mathbf{t}_{-i}}[a_i(t_i, \mathbf{t}_{-i})] \geq \mathbb{E}_{\mathbf{t}_{-i}}[p_i(t_i, \mathbf{t}_{-i})] - \varphi$ for all i and all $t_i \in T$. We also show that the induced interim allocation rule for this merit-with-guarantee mechanism is equal to $-\frac{1}{n}c'_\varphi$. Thus, by Lemma 2 the mechanism achieves the upper bound of the principal's payoff and we conclude that a merit-with-guarantee mechanism is optimal.

We say that the partition $\{T^\ell, T^k, T^m\}$ is *determined by* c_φ if, for all $t \in T^\ell$, $c_\varphi^{ic}(t) \leq \min\{c_\varphi^{aud}(t), c^{allo}(t)\}$, for all $t \in T^k$, $c_\varphi^{aud}(t) \leq \min\{c_\varphi^{ic}(t), c^{allo}(t)\}$, and for all $t \in T^m$, $c^{allo}(t) \leq \min\{c_\varphi^{ic}(t), c_\varphi^{aud}(t)\}$.

A.1 Characterization of BIC mechanisms

Proof of Lemma 1. Let $i \in N$. By definition of BIC a mechanism (p, a) in equation (1) and the fact that $u_i(t_i) > 0$ for all $t_i \in T$ we get that agent i with type $t_i \in T$ has no incentive to deviate if and only if, for all $t'_i \in T$,

$$\mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t_i, \mathbf{t}_{-i}, s)] \geq \mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t'_i, \mathbf{t}_{-i}, s)[1 - a_i(t'_i, \mathbf{t}_{-i}, s)]]. \quad (9)$$

Since $a_i(\mathbf{t}, s) = 1$ only if $p_i(\mathbf{t}, s) = 1$, the right-hand side equals $P(t'_i) - A(t'_i)$. Since (9) is required to hold for all $t_i \in T$, it must in particular hold for the infimum over T . Thus, Definition 1 of BIC is equivalent to the equation in Lemma 1. $\square$

A.2 Feasibility of interim rules

We consider the interim rules induced by the lottery rule p^M and the audit rule a and show that they satisfy the necessary feasibility constraints for interim rules known from the literature.

To begin with, we introduce some notation and definitions that we will use throughout the appendix. For $D_1, D_2 \subseteq [0, 1]$, the disjoint union is defined as $D_1 \sqcup D_2 := \{(t, 1) | t \in D_1\} \cup \{(t, 2) | t \in D_2\}$, and analogously for disjoint unions of n sets. In contrast to the standard union, the disjoint union indexes each element by the set it comes from. For every $E = E_1 \sqcup \dots \sqcup E_n$ with each $E_i \subseteq T$, and every $\mathbf{t} \in \mathbf{T}$, let $I(\mathbf{t}, E) := \{i \in N | t_i \in E_i\}$ denote the set of agents with types in E that are compatible with $\mathbf{t}$. Slightly abusing notation, if $E \subseteq T$, we write $I(\mathbf{t}, E)$ for $I(\mathbf{t}, E \sqcup \dots \sqcup E)$.

In Section 4 we argued that any feasible interim rule must respect the the upper bound c^{allo} and c_φ^{aud} , due to the limited availability of objects and audits. Formally, this can be derived from the following result.

Lemma 3. *An interim allocation rule P is feasible if and only if for all Borel sets $E \subseteq T$,*

$$n \int_E P(t_i) dt_i \leq \int \min\{|I(\mathbf{t}, E)|, m\} d\mathbf{t}. \quad (10)$$

Moreover, if P is non-decreasing, it is enough to check this inequality for sets E of the form $E = [e, 1]$.

The next lemma characterizes interim rules in our setting with type-dependent ex-post constraints. Specifically, we assume that if type profile $\mathbf{t}$ realizes, $h(\mathbf{t})$ objects can be allocated. Moreover, we assume only agents in the set $J(\mathbf{t})$ can get an object. An allocation rule p is *feasible* if, for all $\mathbf{t} \in \mathbf{T}$, $\sum_i \mathbb{E}_s[p_i(\mathbf{t}, s)] \leq h(\mathbf{t})$ and $\mathbb{E}_s[p_i(\mathbf{t}, s)] > 0$ implies $i \in J(\mathbf{t})$.¹⁰ To formulate the lemma we consider asymmetric interim rules $P_i : T_i \rightarrow [0, 1]$ where $P_i(t) := \mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t, \mathbf{t}_{-i}, s)]$.

Lemma 4. *Let $h(\mathbf{t})$ be the total number of objects available, and let $J(\mathbf{t})$ denote the set of agents that can receive an object, which can both depend on the realized type profile. There is a feasible allocation rule $p = (p_i)_{i \in N}$ inducing an interim allocation rule $P = (P_i)_{i \in N}$ if and only if, for all sets $E = \bigsqcup_{i \in N} E_i$, where each E_i is a Borel subset of T_i ,*

$$\sum_i \int_{E_i} P_i(t) dt \leq \int \min\{|J(\mathbf{t}) \cap I(\mathbf{t}, E)|, h(\mathbf{t})\} d\mathbf{t}.$$

Note that for an interim audit rule A to be feasible there must exist an audit rule a with $A(t) = \mathbb{E}_{\mathbf{t}_{-i}, s}[a_i(t, \mathbf{t}_{-i}, s)]$ for each $t \in T$, and $a(\mathbf{t}, s) \leq p(\mathbf{t}, s)$, for the specific p that induces P . We imposed the latter constraint $a(\mathbf{t}, s) \leq p(\mathbf{t}, s)$, since this simplified our characterization of incentive compatibility. However, this additional constraint complicates the characterization of which interim audit rules are feasible given an allocation rule.

To state the characterization, let p be a symmetric allocation rule that is deterministic for almost all $\mathbf{t}$. Define for every $E = E_1 \sqcup \dots \sqcup E_n$ with each $E_i \subseteq T$, and every $\mathbf{t} \in \mathbf{T}$, $J^p(\mathbf{t}, E) := \{i \in N | t_i \in E_i \text{ and } p_i(\mathbf{t}) = 1\}$. Slightly abusing notation, if $E \subseteq T$, we write $J^p(\mathbf{t}, E)$ for $J^p(\mathbf{t}, E \sqcup \dots \sqcup E)$.

Lemma 5. *An interim audit rule A is feasible given a symmetric and deterministic ex-post allocation rule p if and only if for all Borel sets $E \subseteq T$,*

$$n \int_E A(t_i) dt_i \leq \int \min\{|J^p(\mathbf{t}, E)|, k\} d\mathbf{t}. \quad (11)$$

¹⁰With slight abuse of notation, we will drop the realization of the randomization device as an argument for rules that are deterministic for almost every $\mathbf{t}$.

Although the interim rules induced by p^ℓ and a are monotone, it does not suffice to check feasibility constraints in Lemma 4 and 5 only for upper sets. This is because the number of objects available to allocate (or the number of audits available) is effectively dependent on the type profile: how many objects are allocated in the merit stage depends on the type profile, and therefore the number of objects available in the lottery stage; similarly, because our formulation of incentive compatibility assumes that only agents receiving an object can be audited, the number of audits that can be conducted depends on the type profile. If the number of objects is type-dependent, it is not sufficient to check the Border constraints only for upper sets.¹¹

A.3 Properties of the function c_φ and the partition of T determined by c_φ

In this subsection we derive properties of the functions c^{allo} , c_φ^{aud} , and c^{ic} ; we will use these properties later to show that there is a merit-with-guarantee mechanisms for which (7) is binding for all t .

Lemma 6. *The derivatives of c^{allo} and c_φ^{aud} are given by*

$$\frac{d}{dt}(c^{allo}(t)) = -n \sum_{j=0}^{m-1} \binom{n-1}{j} t^{n-1-j} (1-t)^j \quad (12)$$

$$\frac{d}{dt}(c_\varphi^{aud}(t)) = -n \sum_{j=0}^{k-1} \binom{n-1}{j} t^{n-1-j} (1-t)^j - n\varphi. \quad (13)$$

Also, $c^{allo}(0) = c_\varphi^{ic}(0) = m$ and $c^{allo}(1) = c_\varphi^{aud}(1) = 0$. If $\varphi \geq \frac{m-k}{n}$ then $c_\varphi^{aud}(0) \geq c^{allo}(0)$.

Proof of Lemma 6. We will prove (12); an analogous argument can then be used to prove (13).

Let $G(i; n, \rho)$ denote the CDF of the binomial distribution, where i is the number of successes in n independent Bernoulli trials, and ρ is the probability of success. The CDF

¹¹Indeed, consider an example with two agents and two types each, $T = \{\underline{t}, \bar{t}\}$ with $\underline{t} < \bar{t}$. Types are iid and both types are equally likely. Further, all agents can be assigned at most one object at all type profiles but the number of objects available at a given type profile varies. The following table specifies

how many objects are available:

	$\underline{t}_2$	$\bar{t}_2$
$\underline{t}_1$	0	1
$\bar{t}_1$	2	0

Now, consider the following monotone interim rule $P_1(t_1) = P_1(\bar{t}_1) = P_2(\underline{t}_2) = 0.25$ and $P_2(\bar{t}_2) = 0.5$. One can verify that the Border constraints are satisfied for all upper sets, but violated for the set $\{\underline{t}_1, \bar{t}_2\}$. Thus, the interim rule is not feasible.

of the binomial distribution can be written as,

$$G(i; n, \rho) = (n-i) \binom{n}{i} \int_0^{1-\rho} z^{n-i-1} (1-z)^i dz$$

$$\text{and } \frac{\partial G(i; n, \rho)}{\partial \rho} = -(n-i) \binom{n}{i} (1-\rho)^{n-i-1} \rho^i = -n \binom{n-1}{i} (1-\rho)^{n-i-1} \rho^i.$$

Now, using that the probability of success $\rho = 1 - t$ we can express $c^{allo}(t)$ using the binomial distribution as

$$c^{allo}(t) = \sum_{i=1}^m i[G(i; n, 1-t) - G(i-1; n, 1-t)] + m[1 - G(m; n, 1-t)].$$

It then follows that

$$G(i; n, 1-t) = (n-i) \binom{n}{i} \int_0^t z^{n-i-1} (1-z)^i dz$$

$$\frac{\partial G(i; n, 1-t)}{\partial t} = n \binom{n-1}{i} t^{n-i-1} (1-t)^i. \quad (14)$$

To establish equation (12) that gives us the derivative of $c^{allo}(t)$, we use induction over the number of objects m . The claim in (12) is trivially true for $m = 1$.

Induction hypothesis: $\frac{d}{dt}(c^{allo}(t)) = -n \sum_{i=0}^{m-1} \binom{n-1}{i} t^{n-1-i} (1-t)^i$ for all m .

We are going to show that if the induction hypothesis holds for m then it also holds for $m+1$. We will show this by looking at the difference between $c^{allo}(t|m+1 \text{ objects})$ and $c^{allo}(t|m \text{ objects})$ and then differentiating wrt to t using equation (14). That is,

$$\begin{aligned} c^{allo}(t|m+1) - c^{allo}(t|m) &= \sum_{i=1}^{m+1} i[G(i; n, 1-t) - G(i-1; n, 1-t)] + (m+1)[1 - G(m+1; n, 1-t)] \\ &\quad - \left[\sum_{i=1}^m i[G(i; n, 1-t) - G(i-1; n, 1-t)] + m[1 - G(m; n, 1-t)] \right] \\ &= 1 - G(m; n, 1-t) \end{aligned}$$

Now, differentiating with respect to t gives,

$$\begin{aligned} \frac{d}{dt}(c^{allo}(t|m+1) - c^{allo}(t|m)) &= -\frac{\partial G(m; n, 1-t)}{\partial t} = -(n-m) \binom{n}{m} t^{n-m-1} (1-t)^m \\ &= -n \binom{n-1}{m} t^{n-m-1} (1-t)^m. \end{aligned}$$

By the induction hypothesis, $\frac{d}{dt}(c^{allo}(t)) = -n \sum_{i=0}^{m-1} \binom{n-1}{i} t^{n-1-i} (1-t)^i$. Thus, from the above equality, we get that also for $m+1$ objects equation (12) holds. In an analogous argument, also using proof by induction, we can show that equation (13) for $\frac{d}{dt}(c_\varphi^{aud}(t))$ holds.

Note also that, $c^{allo}(0) = m$ and $c_\varphi^{aud}(0) = k + n\varphi$. Thus, $c_\varphi^{aud}(0) \geq c^{allo}(0)$ for $\varphi \geq \frac{m-k}{n}$ as claimed in the lemma. $\square$

Consider a partition $\{T^\ell, T^k, T^m\}$ determined by c_φ . The next lemma establish that for any $\varphi \in [0, \frac{m}{n}]$ such a partition is well defined and coincides with the general form for the partition of the type space $\{T^\ell, T^k, T^m\}$ that is part of a definition of the merit-with-guarantee mechanism.

Lemma 7. *Let $\varphi \in [0, \frac{m}{n}]$. Then, there exist $0 \leq x_1 \leq x_2 \leq x_3 \leq 1$ such that $T^\ell = [0, x_1)$, $T^k = (x_2, x_3)$, and $T^m = (x_1, x_2) \cup (x_3, 1]$ satisfy*

$$\begin{aligned} c_\varphi^{ic}(t) &\leq c_\varphi^{allo}(t) \text{ and } c_\varphi^{ic}(t) \leq c_\varphi^{aud}(t) && \text{for all } t \in T^\ell \\ c_\varphi^{aud}(t) &\leq c_\varphi^{allo}(t) \text{ and } c_\varphi^{aud}(t) \leq c_\varphi^{ic}(t) && \text{for all } t \in T^k \\ c_\varphi^{allo}(t) &\leq c_\varphi^{aud}(t) \text{ and } c_\varphi^{allo}(t) \leq c_\varphi^{ic}(t) && \text{for all } t \in T^m. \end{aligned}$$

Proof. Let $\varphi \in [0, \frac{m}{n}]$. We begin with computing the second derivatives of c_φ^{allo} and c_φ^{aud} . It follows from Lemma 6 that

$$\frac{d}{dt}(c_\varphi^{allo}(t)) = -n \sum_{j=0}^{m-1} \binom{n-1}{j} t^{n-1-j} (1-t)^j = -nG(m-1, n-1, 1-t).$$

Thus,

$$\begin{aligned} \frac{d^2}{dt^2}(c_\varphi^{allo}(t)) &= -n \frac{\partial G(m-1, n-1, 1-t)}{\partial t} = -n(n-m) \binom{n-1}{m-1} t^{n-m-1} (1-t)^{m-1} \\ &= -nm \binom{n-1}{m} t^{n-m-1} (1-t)^{m-1}. \end{aligned}$$

Again, using Lemma 6 for c_φ^{aud} we have:

$$\frac{d}{dt}(c_\varphi^{aud}(t)) = -n \sum_{j=0}^{k-1} \binom{n-1}{j} t^{n-1-j} (1-t)^j - n\varphi = -nG(k-1, n-1, 1-t) - n\varphi.$$

Analogously to the above we obtain that

$$\frac{d^2}{dt^2}(c_\varphi^{aud}(t)) = -nk \binom{n-1}{k} t^{n-k-1} (1-t)^{k-1}.$$

Both second derivatives are negative and we conclude that c_φ^{allo} and c_φ^{aud} are concave functions.

Since c_φ^{ic} is linear, $c_\varphi^{ic}(0) = m = c_\varphi^{allo}(0)$, and $c_\varphi^{ic}(1) \geq 0 = c_\varphi^{allo}(1)$, there is $x_1 \in [0, 1]$ such that:

$$\begin{aligned} c_\varphi^{ic}(t) &\leq c_\varphi^{allo}(t) && \text{for all } t \in [0, x_1] \\ c_\varphi^{allo}(t) &\leq c_\varphi^{ic}(t) && \text{for all } t \in [x_1, 1]. \end{aligned} \tag{15}$$

Also, because c_φ^{aud} is steeper than c_φ^{ic} by (13), there is $x_2 \in [0, 1]$ such that:

$$\begin{aligned} c_\varphi^{ic}(t) &\leq c_\varphi^{aud}(t) && \text{for all } t \in [0, x_2] \\ c_\varphi^{aud}(t) &\leq c_\varphi^{ic}(t) && \text{for all } t \in (x_2, 1]. \end{aligned} \tag{16}$$

Next, let us look at $c^{allo} - c_\varphi^{aud}$. The difference between two concave functions can have arbitrary many crossings. However, by examining the number of zeros of the second derivative of $c^{allo} - c_\varphi^{aud}$ we will show that there can be at most three zeros of $c^{allo} - c_\varphi^{aud}$ on $[0, 1]$. Now, consider the second derivative of $c^{allo} - c_\varphi^{aud}$,

$$\frac{d^2}{dt^2}(c^{allo}(t) - c_\varphi^{aud}(t)) = -nt^{n-m-1}(1-t)^{k-1} \left[m \binom{n-1}{m} (1-t)^{m-k} - k \binom{n-1}{k} t^{m-k} \right].$$

Note that the term in brackets is decreasing in t and, hence, $\frac{d^2}{dt^2}(c^{allo}(t) - c_\varphi^{aud}(t))$ crosses zero at most once. It follows that $\frac{d}{dt}(c^{allo}(t) - c_\varphi^{aud}(t))$ has at most two zeros on $[0, 1]$, which implies that $c^{allo} - c_\varphi^{aud}$ has at most three zeros on $[0, 1]$. Since $c^{allo}(1) = c_\varphi^{aud}(1)$, there are at most two zeros of $c^{allo} - c_\varphi^{aud}$ on $[0, 1]$. Note further that (i): for all $\varphi \in (0, \frac{m}{n}]$, $\frac{d}{dt}(c^{allo}(t)) > \frac{d}{dt}(c_\varphi^{aud}(t))$ for $t = 1$, and (ii): $c^{allo}(t) \leq c_\varphi^{aud}(t)$ for all t close to 1. Thus, $c^{allo}(t) < c_\varphi^{aud}(t)$ for t close to 1, for any $\varphi \in (0, \frac{m}{n}]$. $\square$

A.4 Existence of lottery rule p^ℓ

In this subsection of the appendix we show that there exists a feasible lottery rule p^ℓ with $\mathbb{E}_{\mathbf{t}_{-i}, s}[p^\ell(t_i, \mathbf{t}_{-i}, s)] = \varphi$ for all $t_i \in T^\ell \cup T^k$ and that the allocation rule in the merit-with-guarantee mechanism induces the interim rule $-\frac{1}{n}c'_\varphi$.

Let (p^*, a^*) denote an optimal mechanism and define $\varphi := \inf_t \mathbb{E}[p_i^*(t)]$ and let the partition $\{T^\ell, T^k, T^m\}$ be determined by c_φ . Let p^M denote the corresponding merit-based allocation rule.

Lemma 8. *The interim rule of p^M is $-\frac{1}{n}c'_\varphi - \varphi \mathbf{1}_{T^k \cup T^\ell}$.*

Proof. Note that for type $t_i \in T^m$, the probability that i obtains an object under allocation rule p^M equals the probability that there are at most $m-1$ others with types above t_i . Using the definition of P and Lemma 6, this yields

$$\mathbb{E}_{\mathbf{t}_{-i}}[p_i^M(t_i, \mathbf{t}_{-i})] = \sum_{j=0}^{m-1} \binom{n-1}{j} t_i^{n-1-j} (1-t_i)^j = -\frac{1}{n}c'_\varphi(t_i).$$

If $t_i \in T^k$, the probability that i obtains an object under allocation rule p^M equals the probability that there are at most $k-1$ others with types above t_i . Again using the definition of P and Lemma 6, this yields

$$\mathbb{E}_{\mathbf{t}_{-i}}[p_i^M(t_i, \mathbf{t}_{-i})] = \sum_{j=0}^{k-1} \binom{n-1}{j} t_i^{n-1-j} (1-t_i)^j = -\frac{1}{n}c'_\varphi(t_i) - \varphi.$$

Finally, if $t_i \in T^\ell$ we get

$$\mathbb{E}_{\mathbf{t}_{-i}}[p_i^M(t_i, \mathbf{t}_{-i})] = 0 = -\frac{1}{n}c'_\varphi(t_i) - \varphi.$$

$\square$

Lemma 9. *There is an ex-post allocation rule $p^\ell : \mathbf{T} \times [0, 1] \rightarrow \mathbb{R}^n$ allocating the objects not already allocated under p^M that induces the interim rule $\varphi \mathbf{1}_{T^k \cup T^\ell}$.*

Proof. By Lemma 4, it is sufficient to check the following inequality:

$$\sum_i \int_{E_i} \varphi \mathbf{1}_{T^k \cup T^\ell} dt_i \leq \int \min \left\{ \sum_{i \in I(\mathbf{t}, E_1, \dots, E_n)} 1 - p_i^M(\mathbf{t}), m - \sum_i p_i^M(\mathbf{t}) \right\} d\mathbf{t} \quad (17)$$

Further, it is sufficient to show this inequality for sets satisfying $E_i \subseteq T^\ell \cup T^k$ because the right-hand side is increasing in E_i and the left-hand side only depends on $E_i \cap (T^\ell \cup T^k)$.

We now argue that it is sufficient to check the inequality for sets $(E_1, \dots, E_n)$ where for each j there is $x^j \in [x_2, x_3]$ such that $E_j = [x^j, x_3]$ or $E_j = T^\ell \cup [x^j, x_3]$. A simple rewriting shows that inequality (17) above is satisfied if the following inequality holds for any $(E_1, \dots, E_n)$:

$$\begin{aligned} & \sum_{j \neq i} \int_{E_j} \varphi \mathbf{1}_{t_j \in T^k \cup T^\ell} dt_j \\ & - \int_{\mathbf{T}} \min \left\{ \sum_{j \in I(\mathbf{t}, E_1, \dots, E_n) \setminus \{i\}} 1 - p_j^M(\mathbf{t}), m - \sum_j p_j^M(\mathbf{t}) \right\} d\mathbf{t} \\ & + \int_{E_i} \varphi \mathbf{1}_{t_i \in T^k \cup T^\ell} dt_i \\ & + \int_{E_i \times \mathbf{T}_{-i}} \min \left\{ \sum_{j \in I(\mathbf{t}, E_1, \dots, E_n) \setminus \{i\}} 1 - p_j^M(\mathbf{t}), m - \sum_j p_j^M(\mathbf{t}) \right\} \\ & \quad - \min \left\{ \sum_{j \in I(\mathbf{t}, E_1, \dots, E_n)} 1 - p_j^M(\mathbf{t}), m - \sum_j p_j^M(\mathbf{t}) \right\} d\mathbf{t} \\ & \leq 0. \end{aligned} \quad (18)$$

The fourth and fifth lines can be written as

$$- \int_{E_i \times \mathbf{T}_{-i}} \mathbf{1}_{p_i^M(\mathbf{t})=0 \text{ and } \sum_{j \in I(\mathbf{t}, E_1, \dots, E_n)} 1 - p_j^M(\mathbf{t}) \leq m - \sum_j p_j^M(\mathbf{t})} d\mathbf{t}.$$

After this rewriting, the terms in the LHS of (18) that depend on E_i are line 3 and the rewritten expression above; denote the integrand of the latter by $-g(t_i, \mathbf{t}_{-i})$, where

$$g(t_i, \mathbf{t}_{-i}) := \mathbf{1}_{\left\{ p_i^M(\mathbf{t}) = 0 \text{ and } \sum_{j \in I(\mathbf{t}, E_1, \dots, E_n)} (1 - p_j^M(\mathbf{t})) \leq m - \sum_j p_j^M(\mathbf{t}) \right\}}.$$

To maximize the LHS over the choice of E_i , we argue: (i) Either all or none of T^ℓ should be included in E_i . This is because, for $t_i \in T^\ell$, both the integrand of line 3 (which equals φ) and $g(t_i, \mathbf{t}_{-i})$ are independent of the value of t_i within T^ℓ : the former is constant, and the latter depends only on whether $t_i \in E_i$, not on the specific value of t_i . (ii) Whenever it is optimal to include a type $t_i \in T^k$ in E_i , it is optimal to include all higher types $t'_i \in T^k$. To see this, note that raising t_i to t'_i within T^k does not change the

integrand of line 3 (which remains φ), so it suffices to show that $g(t'_i, \mathbf{t}_{-i}) \leq g(t_i, \mathbf{t}_{-i})$ for each $\mathbf{t}_{-i}$. There are three cases: If $p_i^M(t_i, \mathbf{t}_{-i}) = 1$ then $p_i^M(t'_i, \mathbf{t}_{-i}) = 1$, implying $g(t_i, \mathbf{t}_{-i}) = g(t'_i, \mathbf{t}_{-i}) = 0$; if $p_i^M(t_i, \mathbf{t}_{-i}) = 0$ and $p_i^M(t'_i, \mathbf{t}_{-i}) = 1$ then $g(t_i, \mathbf{t}_{-i}) = 1$ and $g(t'_i, \mathbf{t}_{-i}) = 0$ or $g(t_i, \mathbf{t}_{-i}) = g(t'_i, \mathbf{t}_{-i})$; if $p_i^M(t_i, \mathbf{t}_{-i}) = 0$ and $p_i^M(t'_i, \mathbf{t}_{-i}) = 0$ then the allocation of other agents is not changed and hence $g(t_i, \mathbf{t}_{-i}) = g(t'_i, \mathbf{t}_{-i})$. We conclude that it is sufficient to check the inequality for sets $(E_1, \dots, E_n)$ where for each j there is $x^j \in [x_2, x_3]$ such that $E_j = [x^j, x_3]$ or $E_j = T^\ell \cup [x^j, x_3]$.

We now argue that we can assume, in addition, $E_j = E_i$ for all i, j . To apply Lemma 13, denote the left-hand side of (18) as $f(E_1 \sqcup \dots \sqcup E_n)$. It can be verified that f is symmetric and supermodular. Also, let $G \subseteq \mathcal{B}(H)$ be the collection of subsets of the form $E_1 \sqcup \dots \sqcup E_n$, where for each j , $E_j = [x^j, x_3]$ or $E_j = T^\ell \cup [x^j, x_3]$ for some $x^j \in [x_2, x_3]$. Note that G is closed under permutations, intersections, and unions. Lemma 13 then implies that if $f(E_1 \sqcup \dots \sqcup E_n) > 0$ for $E_1 \sqcup \dots \sqcup E_n \in G$ then there is $E \sqcup \dots \sqcup E \in G$ with $f(E \sqcup \dots \sqcup E) > 0$. Hence, it is sufficient to check the inequality (17) for $E_i = E$ for all i , where $E = [x, x_3]$ or $E = T^\ell \cup [x, x_3]$ for some $x \in [x_2, x_3]$.

Let p_i^{eff} denote the efficient allocation rule, and P^{eff} denote the induced efficient interim rule. For any $x \in [x_2, x_3]$,

$$n \int_x^{x_3} P(t_i) dt_i = c^{aud}(x) - c^{allo}(x_3) \leq c^{allo}(x) - c^{allo}(x_3) = n \int_x^{x_3} P^{eff}(t_i) dt_i.$$

Also,

$$n \int_{T^\ell} P(t_i) dt_i = c^{allo}(0) - c^{allo}(x_1) = n \int_{T^\ell} P^{eff}(t_i) dt_i.$$

Because $p_i^{eff}(\mathbf{t}) \leq 1$ for all i and $\sum_i p_i^{eff}(\mathbf{t}) \leq m$, we get that for any set E ,

$$n \int_E P^{eff}(t_i) dt_i = \int \sum_{i \in I(\mathbf{t}, E)} p_i^{eff}(\mathbf{t}) d\mathbf{t} \leq \int \min \left\{ |I(\mathbf{t}, E)|, m - \sum_{i \notin I(\mathbf{t}, E)} p_i^{eff}(\mathbf{t}) \right\} d\mathbf{t}.$$

Putting these inequalities together and noting that $p_i^M(\mathbf{t}) \leq p_i^{eff}(\mathbf{t})$ for all $\mathbf{t}$ and i , we obtain that for any set E of the form $E = [x, x_3]$ or $E = T^\ell \cup [x, x_3]$ for $x \in [x_2, x_3]$,

$$n \int_E P(t_i) dt_i \leq \int \min \left\{ |I(\mathbf{t}, E)|, m - \sum_{j \notin I(\mathbf{t}, E)} p_j^M(\mathbf{t}) \right\} d\mathbf{t}.$$

Subtracting $\sum_{j \in I(\mathbf{t}, E)} p_j^M(\mathbf{t})$ from both sides, we obtain

$$n \int_E \varphi dt_i \leq \int \min \left\{ \sum_{j \in I(\mathbf{t}, E)} 1 - p_j^M(\mathbf{t}), m - \sum_j p_j^M(\mathbf{t}) \right\} d\mathbf{t}.$$

Therefore, the inequality in Lemma 4 is satisfied. $\square$

We let $p = p^\ell + p^M$, denote by P its interim allocation rule. Note that $P(t) = -\frac{1}{n} c'_\varphi(t)$ for all t .

A.5 Feasibility of the verification rule

Define $A := P - \varphi$. We will show that A is feasible given $p = p^\ell + p^M$ by proving the stronger result that A is feasible given p^M .

Lemma 10. *If*

$$n \int_E A(t_i) dt_i \leq \int \min\{|J^{p^M}(\mathbf{t}, E)|, k\} d\mathbf{t} \quad (19)$$

holds for all sets of the form $E = ([x, x_3] \cup [x', 1])$ where $x \in [x_1, x_2]$ and $x' \in [x_3, 1]$ then it is satisfied for all sets $E \subseteq T$.

Proof. For any collection $(E_1, \dots, E_n)$, where each $E_i \subseteq T$, define the function H by

$$\begin{aligned} H(E_1, \dots, E_n) &:= \sum_j \int_{E_j \times T_{-j}} (\mathbf{1}_{t_j \in T^m} [p_j^M(t_j, \mathbf{t}_{-j}) - \varphi] + \mathbf{1}_{t_j \in T^k} p_j^M(t_j, \mathbf{t}_{-j})) d\mathbf{t} \\ &\quad - \int_T \min \left\{ k, \sum_j \mathbf{1}_{t_j \in E_j} p_j^M(t_j, \mathbf{t}_{-j}) \right\} d\mathbf{t}. \end{aligned}$$

Note that, for any $E \subseteq T$, $H(E, \dots, E)$ equals the left-hand side of (19) minus the right-hand side of (19). Therefore, it is sufficient to show $H(E, \dots, E) \leq 0$.

Fixing an arbitrary agent i , we can rewrite $H(E_1, \dots, E_n)$ as

$$\begin{aligned} &\sum_{j \neq i} \int_{E_j \times T_{-j}} (\mathbf{1}_{t_j \in T^m} [p_j^M(t_j, \mathbf{t}_{-j}) - \varphi] + \mathbf{1}_{t_j \in T^k} p_j^M(t_j, \mathbf{t}_{-j})) d\mathbf{t} \\ &- \int_T \min \left\{ k, \sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j^M(t_j, \mathbf{t}_{-j}) \right\} d\mathbf{t} \\ &+ \int_{E_i \times T_{-i}} (\mathbf{1}_{t_i \in T^m} [p_i^M(t_i, \mathbf{t}_{-i}) - \varphi] + \mathbf{1}_{t_i \in T^k} p_i^M(t_i, \mathbf{t}_{-i})) d\mathbf{t} \\ &+ \int_{E_i \times T_{-i}} \left(\min \left\{ k, \sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j^M(t_j, \mathbf{t}_{-j}) \right\} - \min \left\{ k, \sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j^M(t_j, \mathbf{t}_{-j}) + p_i^M(t_i, \mathbf{t}_{-i}) \right\} \right) d\mathbf{t} \end{aligned}$$

Observe that only the third and fourth lines of the left-hand side depend on E_i . It can be shown that, for any $E_1, \dots, E_{i-1}, E_{i+1}, \dots, E_n$, $H(E_1, \dots, E_{i-1}, \cdot, E_{i+1}, \dots, E_n)$ is maximized by setting¹²

$$E_i = T^k \cup \left\{ t_i \in T^m : \int_{\{t_{-i} \in T_{-i} : \sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j^M(t_j, \mathbf{t}_{-i}) \geq k\}} p_i^M(t_i, \mathbf{t}_{-i}) d\mathbf{t}_{-i} \geq \varphi \right\}. \quad (20)$$

We are now going to argue that it is maximized by a set of the form $E_i = ([x, x_3] \cup [x', 1])$ where $x \in [x_1, x_2]$ and $x' \in [x_3, 1]$.

¹²This follows because the integrand in the fourth line equals 0 or -1, and -1 exactly if $\sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j^M(t_i, \mathbf{t}_{-i}) < k$ and $p_i^M(\mathbf{t}) = 1$.

Fix arbitrary $\mathbf{t}_{-i}$ and $t_i \in T^m$ such that $t_i < x_2$ and $p_i^M(t_i, \mathbf{t}_{-i}) = 1$. Choose arbitrary $t'_i \in T^m$ such that $x_2 \geq t'_i > t_i$ and $t'_i \neq t_j$ for all $j \neq i$. Then $p_i^M(t'_i, \mathbf{t}_{-i}) = 1$ and $p_j^M(t_i, \mathbf{t}_{-i}) = p_j^M(t'_i, \mathbf{t}_{-i})$ for all $j \neq i$ since t_i is among the m highest types and j with type t_j gets an object if and only if either t_j is among the m highest types and in T^m or if t_j is among the k highest types and in T^k (in which case he remains among the k highest types if i 's type increases to t'_i since $t'_i \leq x_2$). Therefore, we conclude that the integral in (20) is non-decreasing in t_i on $[x_1, x_2]$.

Similarly, fix arbitrary $\mathbf{t}_{-i}$ and $t_i \in T^m$ such that $t_i \geq x_3$ and $p_i^M(t_i, \mathbf{t}_{-i}) = 1$. Choose arbitrary $t'_i \in T^m$ such that $t'_i > t_i$ and $t'_i \neq t_j$ for all $j \neq i$. Then $p_i^M(t'_i, \mathbf{t}_{-i}) = 1$ and $p_j^M(t_i, \mathbf{t}_{-i}) = p_j^M(t'_i, \mathbf{t}_{-i})$ for all $j \neq i$ since i is among the m highest types and j gets an object if and only if he is either among the m highest types and has a type in T^m or he is among the k highest types and has a type in T^k (in which case he remains among the k highest types since $x_3 \leq t_i < t'_i$). We conclude that the integral in (20) is non-decreasing in t_i on $[x_3, 1]$. It follows that $H(E_1, \dots, E_{i-1}, \cdot, E_{i+1}, \dots, E_n)$ is maximized by a set of the form $E_i = ([x, x_3] \cup [x', 1])$ where $x \in [x_1, x_2]$ and $x' \in [x_3, 1]$.

Since H is symmetric and supermodular, Lemma 13 implies that there are $x \in [x_1, x_2]$ and $x' \in [x_3, 1]$ such that H is maximized by $(E', \dots, E')$, where $E' = ([x, x_3] \cup [x', 1])$. Since the left-hand side of (19) minus the right-hand side of that inequality evaluated at E' is nonpositive by assumption and equal to $H(E', \dots, E')$ by construction, we conclude that, for any $E \subseteq T$, $H(E, \dots, E) \leq 0$. $\square$

Lemma 11. Fix arbitrary $x \in [x_1, x_2]$ and $x' \in [x_3, 1]$, and let $E = [x, x_3] \cup [x', 1]$. Then,

$$n \int_E A(t_i) dt_i \leq \int \min\{|J^{p^M}(\mathbf{t}, E)|, k\} d\mathbf{t}.$$

Proof. Define the allocation rule $p^k : \mathbf{T} \rightarrow [0, 1]$ as follows: $p_j^k(\mathbf{t}) = 1$ if $t_k \neq t_l$ for all $k \neq l$, $t_j \in T^m \cup T^k$ and j 's type is among the k highest ones at $\mathbf{t}$; $p_j^k(\mathbf{t}) = 0$ otherwise. Then

$$\begin{aligned} \int_x^{x_2} P(t_i) dt_i &= -\frac{1}{n} \int_x^{x_2} \frac{d}{dt_i} (c^{allo}(t_i)) dt_i = -\frac{1}{n} [c^{allo}(x_2) - c^{allo}(x)] \\ &\leq -\frac{1}{n} [c_\varphi^{aud}(x_2) - c_\varphi^{aud}(x)] = -\frac{1}{n} \int_x^{x_2} \frac{d}{dt_i} (c_\varphi^{aud}(t_i)) dt_i \\ &= \int_x^{x_2} \left(\sum_{j=0}^{k-1} \binom{n-1}{j} t_i^{n-1-j} (1-t_i)^j + \varphi \right) dt_i \\ &= \int_x^{x_2} \left(\int_{T_{-i}} p_i^k(t_i, \mathbf{t}_{-i}) d\mathbf{t}_{-i} + \varphi \right) dt_i, \end{aligned}$$

where the first equality follows from Lemma 8, the inequality follows from Lemma 7, and the equality in the third line follows from Lemma 6.

Similarly, using the definition of P , Lemma 7 and Lemma 6 it follows that,

$$\begin{aligned}\int_{x_2}^{x_3} P(t_i) dt_i &= -\frac{1}{n} \int_{x_2}^{x_3} \frac{d}{dt_i} (c_\varphi^{aud}(t_i)) dt_i \\ &= \int_{x_2}^{x_3} \left(\sum_{j=0}^{k-1} \binom{n-1}{j} t_i^{n-1-j} (1-t_i)^j + \varphi \right) dt_i \\ &= \int_{x_2}^{x_3} \left(\int_{\mathbf{T}_{-i}} p_i^k(t_i, \mathbf{t}_{-i}) d\mathbf{t}_{-i} + \varphi \right) dt_i.\end{aligned}$$

Using the definition of P and Lemma 7, we obtain

$$\begin{aligned}\int_{x'}^1 P(t_i) dt_i &= -\frac{1}{n} \int_{x'}^1 \frac{d}{dt_i} (c^{allo}(t_i)) dt_i = -\frac{1}{n} [c^{allo}(1) - c^{allo}(x')] \\ &\leq -\frac{1}{n} \int_{x'}^1 \frac{d}{dt_i} (c_\varphi^{aud}(t_i)) dt_i = \int_{x'}^1 \left(\int_{\mathbf{T}_{-i}} p_i^k(t_i, \mathbf{t}_{-i}) d\mathbf{t}_{-i} + \varphi \right) dt_i.\end{aligned}$$

The inequality follows since $c^{allo}(x') \leq c_\varphi^{aud}(x')$ and $c^{allo}(1) = c_\varphi^{aud}(1)$, and the last equality follows from Lemma 6.

Hence,

$$\begin{aligned}n \int_E A(t_i) dt_i &= n \int_E P(t_i) - \varphi dt_i \\ &\leq \sum_i \int_E \int_{\mathbf{T}_{-i}} p_i^k(t_i, \mathbf{t}_{-i}) d\mathbf{t}_{-i} dt_i \\ &= \sum_i \int_E \int_{\mathbf{T}_{-i}} 1_{i \in J^{p^k}(t_i, \mathbf{t}_{-i}, E)} d\mathbf{t}_{-i} dt_i \\ &= \int_{\mathbf{T}} \sum_i 1_{i \in J^{p^k}(t_i, \mathbf{t}_{-i}, E)} d\mathbf{t} \\ &= \int_{\mathbf{T}} |J^{p^k}(t_i, \mathbf{t}_{-i}, E)| d\mathbf{t} \\ &\leq \int_{\mathbf{T}} \min \left\{ k, |J^{p^M}(t_i, \mathbf{t}_{-i}, E)| \right\} d\mathbf{t} \\ &= \int_{\mathbf{T}} \min \{ |J^{p^M}(t_i, \mathbf{t}_{-i}, E)|, k \} d\mathbf{t},\end{aligned}$$

where the second inequality uses that $|J^{p^k}(\mathbf{t}, E)| \leq k$ and $p_j^k(\mathbf{t}) \leq p_j^M(\mathbf{t})$ for each $\mathbf{t} \in \mathbf{T}$ and $j \in N$. $\square$

Lemma 12. *The interim audit rule A is feasible given p^M .*

Proof. Note that if A is feasible given p^M , it is also feasible given p because, for any $E \subset [0, 1]$ and $\mathbf{t} \in \mathbf{T}$, if $i \in J^{p^M}(\mathbf{t}, E)$ then $i \in J^p(\mathbf{t}, E)$ and therefore

$$\int \min \{ |J^{p^M}(\mathbf{t}, E)|, k \} d\mathbf{t} \leq \int \min \{ |(J^p(\mathbf{t}, E))|, k \} d\mathbf{t}.$$

Moreover, if a is feasible given p^M then it is also feasible given p , since $p_i^M(t_i, \mathbf{t}_{-i}, s) \leq p_i(t_i, \mathbf{t}_{-i}, s)$. Thus, it suffices to show that the verification rule $A(t_i) = P(t_i) - \varphi$ is feasible given p^M . This follows from Lemmas 5, 10, and 11. $\square$

A.6 Completing the proof of Theorem 1

By Lemma 12, there is an audit rule a that is feasible given p with interim rule A .

The mechanism (p, a) satisfies $\mathbb{E}_{\mathbf{t}_{-i}, s}[a_i(t_i, \mathbf{t}_{-i}, s)] \geq \mathbb{E}_{\mathbf{t}_{-i}, s}[p_i(t_i, \mathbf{t}_{-i}, s)] - \varphi$ and is therefore BIC. Because the allocation rule has the structure of a merit-with-guarantee mechanism by construction, (p, a) is therefore a merit-with-guarantee mechanism.

To see that (p, a) is optimal, observe that by Lemmas 8 and 9, the interim allocation rule of p satisfies $P(t) = -\frac{1}{n}c'_\varphi(t)$ for all t and therefore $\int_t^1 P(s) ds = \frac{1}{n}c_\varphi(t)$ (recall that $c_\varphi(1) = 0$). For any other mechanism $(\tilde{p}, \tilde{a})$ with $\inf_t \mathbb{E}_{\mathbf{t}_{-i}, s}[\tilde{p}_i(t, \mathbf{t}_{-i}, s)] \geq \varphi$, it follows from Lemma 2 that

$$\mathbb{E}_{\mathbf{t}, s}[\sum_i \tilde{p}_i(\mathbf{t}, s)v(t_i)] \leq \int_0^1 c_\varphi(t) dv(t)$$

and the mechanism (p, a) achieves equality. Therefore, it is optimal. It follows from Proposition 1 that $0 < x_1 \leq x_2 < x_3 < 1$.

Now let $(\tilde{p}, \tilde{a})$ be an optimal mechanism. Let $\tilde{\varphi} := \inf_{t'} \mathbb{E}_{\mathbf{t}_{-i}, s}[\tilde{p}_i(t'_i, \mathbf{t}_{-i}, s)]$. It then follows from Lemma 2 that for any optimal mechanism, (7) must hold as an equality for all t . It follows that $\tilde{p}$ can be decomposed as described in Definition 2. The properties of $\tilde{a}$ follow from incentive compatibility of $(\tilde{p}, \tilde{a})$. $\square$

B Additional results

B.1 Proof of Proposition 1

The principal's payoff per agent from using a merit-with-guarantee rule with a parameter φ is,

$$\begin{aligned} U(\varphi) &= \int_0^{x_1(\varphi)} -\frac{1}{n}c_\varphi^{ic'}(t)v(t) dt + \int_{x_1(\varphi)}^{x_2(\varphi)} -\frac{1}{n}c_\varphi^{allo'}(t)v(t) dt \\ &\quad + \int_{x_2(\varphi)}^{x_3(\varphi)} -\frac{1}{n}c_\varphi^{aud'}(t)v(t) dt + \int_{x_3(\varphi)}^1 -\frac{1}{n}c_\varphi^{allo'}(t)v(t) dt \end{aligned}$$

Now define three implicit functions:

$$\begin{aligned} h_1(x_1(\varphi), \varphi) &:= c_\varphi^{allo}(x_1(\varphi)) - c_\varphi^{ic}(x_1(\varphi)) \\ h_2(x_2(\varphi), \varphi) &:= c_\varphi^{aud}(x_2(\varphi)) - c_\varphi^{allo}(x_2(\varphi)) \\ h_3(x_3(\varphi), \varphi) &:= c_\varphi^{allo}(x_3(\varphi)) - c_\varphi^{aud}(x_3(\varphi)) \end{aligned}$$

Note that $h_i(x_i(\varphi), \varphi) = 0$ for $i = 1, 2, 3$. By the implicit function theorem, we have

$$\begin{aligned} x_1'(\varphi) &= -\frac{\frac{\partial(c^{allo}(x_1(\varphi)) - c_\varphi^{ic}(x_1(\varphi)))}{\partial\varphi}}{c^{allo'}(x_1(\varphi)) - c_\varphi^{ic'}(x_1(\varphi))} = -\frac{nx_1(\varphi)}{c^{allo'}(x_1(\varphi)) - c_\varphi^{ic'}(x_1(\varphi))} \\ x_2'(\varphi) &= -\frac{\frac{\partial(c_\varphi^{aud}(x_2(\varphi)) - c^{allo}(x_2(\varphi)))}{\partial\varphi}}{c_\varphi^{aud'}(x_2(\varphi)) - c^{allo'}(x_2(\varphi))} = -\frac{n(1 - x_2(\varphi))}{c_\varphi^{aud'}(x_2(\varphi)) - c^{allo'}(x_2(\varphi))} \\ x_3'(\varphi) &= -\frac{\frac{\partial(c^{allo}(x_3(\varphi)) - c_\varphi^{aud}(x_3(\varphi)))}{\partial\varphi}}{c^{allo'}(x_3(\varphi)) - c_\varphi^{aud'}(x_3(\varphi))} = -\frac{-n(1 - x_3(\varphi))}{c^{allo'}(x_3(\varphi)) - c_\varphi^{aud'}(x_3(\varphi))} \end{aligned}$$

Now let us compute $U'(\varphi)$, using Leibniz integral rule,

$$\begin{aligned} U'(\varphi) &= \frac{1}{n} \left[c^{allo'}(x_1(\varphi)) - c_\varphi^{ic'}(x_1(\varphi)) \right] v(x_1(\varphi)) x_1'(\varphi) + \frac{1}{n} \left[c_\varphi^{aud'}(x_2(\varphi)) - c^{allo'}(x_2(\varphi)) \right] v(x_2(\varphi)) x_2'(\varphi) \\ &\quad + \frac{1}{n} \left[c^{allo'}(x_3(\varphi)) - c_\varphi^{aud'}(x_3(\varphi)) \right] v(x_3(\varphi)) x_3'(\varphi) + \int_0^{x_1(\varphi)} v(t) dt + \int_{x_2(\varphi)}^{x_3(\varphi)} v(t) dt \\ &= -x_1(\varphi)v(x_1(\varphi)) - (1 - x_2(\varphi))v(x_2(\varphi)) + (1 - x_3(\varphi))v(x_3(\varphi)) + \int_0^{x_1(\varphi)} v(t) dt + \int_{x_2(\varphi)}^{x_3(\varphi)} v(t) dt \end{aligned}$$

Setting $U'(\varphi) = 0$ yields

$$\int_0^{x_1(\varphi)} v(t) dt + \int_{x_2(\varphi)}^{x_3(\varphi)} v(t) dt = x_1(\varphi)v(x_1(\varphi)) + (1 - x_2(\varphi))v(x_2(\varphi)) - (1 - x_3(\varphi))v(x_3(\varphi)),$$

just as stated in Proposition 1.

To show that T^k and T^ℓ are nonempty, observe first that $c_\varphi^{ic}(t) < c^{allo}(t)$ for some $t \in (0, 1)$ because $c_\varphi^{ic}(0) = c^{allo}(0)$, c_φ^{ic} has a constant and strictly negative slope, but the slope of c^{allo} goes to zero for t small enough. But if T^ℓ is empty (that is, $x_1(\varphi) = 0$) then $c_\varphi^{aud}(t) < c_\varphi^{ic}(t) < c^{allo}(t)$ for some $t \in (0, 1)$ and hence T^k is nonempty and $x_2(\varphi) < x_3(\varphi)$. Hence, it is not possible that $x_1(\varphi) = 0$ and $x_2(\varphi) = x_3(\varphi)$.

If $x_2(\varphi) = x_3(\varphi)$ then (8) simplifies to

$$x_1(\varphi)v(x_1(\varphi)) = \int_0^{x_1(\varphi)} v(t) dt.$$

which only holds if $x_1(\varphi) = 0$. Since $x_2(\varphi) = x_3(\varphi)$ and $x_1(\varphi) = 0$ are not jointly possible, we conclude that for the optimal mechanism, $x_2(\varphi) < x_3(\varphi)$ and T^k is nonempty.

If $x_1(\varphi) = 0$ then (8) simplifies to

$$v(x_2(\varphi))[1 - x_2(\varphi)] = v(x_3(\varphi))[1 - x_3(\varphi)] + \int_{x_2(\varphi)}^{x_3(\varphi)} v(t) dt.$$

Integrating by parts and simplifying, this yields

$$v(x_3(\varphi)) - v(x_2(\varphi)) = \int_{x_2(\varphi)}^{x_3(\varphi)} t dv(t).$$

Since $v(x_3(\varphi)) - v(x_2(\varphi)) = \int_{x_2(\varphi)}^{x_3(\varphi)} 1 dv(t)$, this holds only if $x_2(\varphi) = x_3(\varphi)$. We conclude that $x_1(\varphi) > 0$ and T^ℓ is nonempty. $\square$

B.2 Proof of Proposition 2

Step 1: Convergence of $\frac{1}{n}c_\varphi$ for fixed φ

For any fixed value of φ , we establish the pointwise convergence of each component of c_φ .

For $\frac{1}{n}c^{\text{allo}}(t)$: Recall that $-\frac{d}{dt}c^{\text{allo}}(t) = \sum_{j=0}^{m-1} \binom{n-1}{j} t^{n-1-j} (1-t)^j$. This sum represents the probability of at most $m-1$ successes in $n-1$ independent Bernoulli trials with success probability $1-t$. For any fixed t , as $n \rightarrow \infty$ with $m_n/n \rightarrow \alpha$:

- If $t < 1 - \alpha$: The expected number of successes $(n-1)(1-t)$ exceeds $m-1$ for n large enough, so by the Strong Law of Large Numbers, this probability converges to 0.
- If $t > 1 - \alpha$: The expected number of successes $(n-1)(1-t)$ is less than $m-1$ for n large enough, so this probability converges to 1.

Therefore:

$$-\frac{1}{n} \frac{d}{dt} c^{\text{allo}}(t) \rightarrow \begin{cases} 0 & \text{if } t < 1 - \alpha \\ 1 & \text{if } t > 1 - \alpha. \end{cases}$$

Since $\frac{1}{n}c^{\text{allo}}(0) = \frac{m_n}{n} \rightarrow \alpha$ and the derivative converges, the dominated convergence theorem yields:

$$\frac{1}{n}c^{\text{allo}}(t) \rightarrow \begin{cases} \alpha & \text{if } t < 1 - \alpha \\ \alpha - (t - 1 + \alpha) & \text{if } t > 1 - \alpha. \end{cases}$$

For $\frac{1}{n}c_\varphi^{\text{aud}}(t)$: By analogous arguments using the fact that $k_n/n \rightarrow \beta$:

$$\frac{1}{n}c_\varphi^{\text{aud}}(t) \rightarrow \begin{cases} \beta + \varphi(1-t) & \text{if } t < 1 - \beta \\ (1 + \varphi)(1-t) & \text{if } t > 1 - \beta. \end{cases}$$

For $\frac{1}{n}c_\varphi^{\text{ic}}(t)$: This function is exact for all n :

$$\frac{1}{n}c_\varphi^{\text{ic}}(t) = \alpha - t\varphi$$

Step 2: Convergence of the optimal guarantee φ_n

Let φ_n denote the optimal guarantee in a market of size n . We claim that $\varphi_n \rightarrow \alpha - \beta$.

First, from the optimality conditions on φ_n , see Proposition 1, we know that $\varphi_n \geq \frac{m_n - k_n}{n}$ for all n . Since $\frac{m_n - k_n}{n} \rightarrow \alpha - \beta$, we have $\liminf_{n \rightarrow \infty} \varphi_n \geq \alpha - \beta$. On the other hand, suppose towards contradiction that

$$\limsup_{n \rightarrow \infty} \varphi_n > \alpha - \beta.$$

Then there exists $\epsilon > 0$ and a subsequence with $\varphi_{n_k} \geq \alpha - \beta + \epsilon$ for all k . For k large enough, $\frac{1}{n_k} c_{\varphi_{n_k}}^{\text{aud}}(t) > \frac{1}{n_k} c_{\varphi_{n_k}}^{\text{ic}}(t)$ for all $t \in [0, 1 - \beta]$ because $\frac{1}{n_k} c_{\varphi_{n_k}}^{\text{aud}}(0) > \frac{1}{n_k} c_{\varphi_{n_k}}^{\text{ic}}(0)$ and both functions the same slope on $[0, 1 - \beta]$.

Similarly, for k large enough, $\frac{1}{n_k} c_{\varphi_{n_k}}^{\text{aud}}(t) > \frac{1}{n_k} c^{\text{allo}}(t)$ for all $t \in (1 - \beta, 1)$ because both functions coincide for $t = 1$ and the derivative of the first function converges on $(1 - \beta, 1]$ to $-1 - \varphi_{n_k}$ and the derivative of the second function converges to -1 .

Together this implies that for k large enough $c_{\varphi_{n_k}}^{\text{aud}}$ is never the binding constraint in equation (7), and therefore T^k is empty contradicting Proposition 1, and we conclude that $\varphi_n \rightarrow \alpha - \beta$.

Step 3: Convergence of the envelope function $\frac{1}{n} c_{\varphi_n}$

With $\varphi_n \rightarrow \alpha - \beta$, the envelope function $\frac{1}{n} c_{\varphi_n}(t) = \frac{1}{n} \min\{c^{\text{allo}}(t), c_{\varphi_n}^{\text{aud}}(t), c_{\varphi_n}^{\text{ic}}(t)\}$ converges as follows.

For $t < \frac{1-\alpha}{1-\alpha+\beta}$, we have

$$\lim \frac{1}{n} c^{\text{allo}}(t) > \lim \frac{1}{n} c_{\varphi_n}^{\text{aud}}(t) = \lim \frac{1}{n} c_{\varphi_n}^{\text{ic}}(t).$$

And for $t > \frac{1-\alpha}{1-\alpha+\beta}$, we have

$$\lim \frac{1}{n} c^{\text{allo}}(t) < \lim \frac{1}{n} c_{\varphi_n}^{\text{aud}}(t) \leq \lim \frac{1}{n} c_{\varphi_n}^{\text{ic}}(t).$$

Therefore:

$$\frac{1}{n} c_{\varphi_n}(t) \rightarrow \begin{cases} \alpha - (\alpha - \beta)t & \text{if } t < \frac{1-\alpha}{1-\alpha+\beta} \\ \alpha - (t - 1 + \alpha) & \text{if } t > \frac{1-\alpha}{1-\alpha+\beta}. \end{cases}$$

Step 4: Convergence of the allocation rule P_n

Since $\frac{1}{n} c_{\varphi_n}$ is concave for each n and the limiting function is differentiable except at $t = \frac{1-\alpha}{1-\alpha+\beta}$, standard results on convergence of concave functions (Theorem D.6.2.7 in (Hiriart-Urruty and Lemaréchal 2004)) imply that the derivatives converge:

$$\frac{1}{n} c'_{\varphi_n}(t) \rightarrow \begin{cases} -(\alpha - \beta) & \text{if } t < \frac{1-\alpha}{1-\alpha+\beta} \\ -1 & \text{if } t > \frac{1-\alpha}{1-\alpha+\beta}. \end{cases}$$

By the optimality condition $P_n(t) = -\frac{1}{n} c'_{\varphi_n}(F(t))$, we obtain:

$$P_n(t) \rightarrow \begin{cases} \alpha - \beta & \text{if } F(t) < \frac{1-\alpha}{1-\alpha+\beta} \\ 1 & \text{if } F(t) > \frac{1-\alpha}{1-\alpha+\beta}. \end{cases}$$

This completes the proof. □

C When is T^m an interval in the optimal mechanism?

If we fix n, m, k , then we can check for a given guarantee φ computationally that T^m consists of only one interval. However, since any value of $\varphi \in [\frac{m-k}{n}, \frac{m}{n}]$ can be optimal for some valuation function v , we can verify directly that the optimal T^m is an interval for given n, m, k . Therefore, we develop theoretical arguments that allow us to check a simple sufficient condition which is independent of v and which implies that T^m is an interval.

Proposition 3. *Let $x_b \in (0, 1]$ satisfy $c^{allo}(x_b) = c_{\frac{m-k}{n}}^{ic}(x_b)$ and define*

$$\varphi_b := \frac{c^{allo}(x_b) - \sum_{i=1}^n \min\{i, k\}(1-x_b)^i x_b^{n-i}}{n(1-x_b)}.$$

If $\varphi_b > m/n$ or $c_{\varphi_b}^{aud'}(x_b) > c^{allo'}(x_b)$, then the optimal T^m is an interval.

Proof. Recall that T^m is the union of at most two intervals; moreover, in an optimal merit-with-guarantee mechanism, there is φ such that $T^m = \{t \in [0, 1] : c^{allo}(t) \leq \min\{c_{\varphi}^{aud}(t), c_{\varphi}^{ic}(t)\}\}$. Now define $x_b \in (0, 1]$ to be the unique solution to $c^{allo}(x_b) = c_{\frac{m-k}{n}}^{ic}(x_b)$ and observe that $c_{\varphi}^{ic}(t) \leq c^{allo}(t)$ for all $t \leq x_b$ and $\varphi \in [\frac{m-k}{n}, \frac{m}{n}]$. Also define $\varphi^* := \frac{c^{allo}(x_b) - \sum_{i=1}^n \min\{i, k\}(1-x_b)^i x_b^{n-i}}{n(1-x_b)}$ and note that $c^{allo}(x_b) = c_{\varphi^*}^{aud}(x_b)$.

Case 1: $\varphi^* > m/n$.

In this case, $c^{allo}(x_b) > c_{\varphi}^{aud}(x_b)$ for all $\varphi \in [\frac{m-k}{n}, \frac{m}{n}]$. Since $c^{allo}(0) \leq c_{\varphi}^{aud}(0)$ and c^{allo} and c_{φ}^{aud} intersect at most twice in $[0, 1]$ (Lemma 7), this implies that T^m is an interval for all $\varphi \in [\frac{m-k}{n}, \frac{m}{n}]$.

Case 2: $\varphi^* \leq m/n$.

Observe that if $c_{\varphi^*}^{aud'}(x_b) > c^{allo'}(x_b)$ then the function $c_{\varphi^*}^{aud}$ crosses c^{allo} from below at x_b (as we approach x_b from the left). Since $c_{\varphi^*}^{aud}(0) \geq c^{allo}(0)$ and the two functions cross at most twice in $[0, 1]$ (Lemma 7), this implies that T^m is an interval. $\square$

Proof of Corollary 1. We verified computationally that the sufficient condition in Proposition 3 is satisfied for all parameter combinations (n, m, k) with $3 \leq n \leq 100$, $2 \leq m < n$, and $1 \leq k < m$. This covers a total of 156,849 cases. For each case, we computed x_b as the solution to $c^{allo}(x_b) = c_{(m-k)/n}^{ic}(x_b)$ and the corresponding

$$\varphi^* = \frac{c^{allo}(x_b) - \sum_{i=1}^n \min\{i, k\} \binom{n}{i} (1-x_b)^i x_b^{n-i}}{n(1-x_b)}.$$

Among all cases, 9,524 (6.1%) satisfy $\varphi^* > m/n$ (Case 1 of the proof), and the remaining 147,325 (93.9%) satisfy $c_{\varphi^*}^{aud'}(x_b) > c^{allo'}(x_b)$ (Case 2). In both cases, the sufficient condition guarantees that T^m is an interval. To ensure numerical reliability, we verified the derivative inequality using interval arithmetic with 50 decimal digits of precision. $\square$

D Generalizations of Border's theorem

In this subsection, we collect all proofs on characterizations of feasible interim rules.

We begin with the result on the feasibility of interim allocation rules. This result is derived from a generalization of Border (1991).

Proof of Lemma 3. This follows immediately from Corollary 4(ii) in Che et al. (2013). $\square$

The next proof extends the characterization of feasible interim rules to settings with type-dependent ex-post constraints. This characterization is derived from a recent result in Valenzuela-Stookey (2024).

Proof of Lemma 4. We first prove the result for finite type spaces and then extend the result to continuous type spaces via an approximation argument. To this end, suppose first that T is finite.

Now we can use Proposition 6 in Valenzuela-Stookey (2024) to establish the statement in Lemma 4 when the type space is finite.

Finally, the approximation argument used to prove Theorem 5 in Che et al. (2013) establishes the conclusion for our setting with infinitely many types. $\square$

We can now use the characterization result on feasible interim allocation rules with type-dependent ex-post constraints to prove the result on feasibility of interim audit rules.

Proof of Lemma 5. By Lemma 4 it follows that there is a feasible audit rule $(a_i(\mathbf{t}))_{i \in N}$ inducing A if and only if, for all sets $E = \bigsqcup_{i \in N} E_i$, where each E_i is a Borel subset of T , and possible $E_i \neq E_j$,

$$\sum_i \int_{E_i} A(t) dt \leq \int \min\{|J^p(\mathbf{t}, E)|, k\} d\mathbf{t}.$$

We can rewrite this as

$$\begin{aligned} & \sum_{j \neq i} \int_{E_j} A(t) dt - \int_T \min \left\{ \sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j(\mathbf{t}), k \right\} d\mathbf{t} + \int_{E_i} A(t) dt \\ & + \int_{E_i} \int_{T-i} \min \left\{ \sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j(\mathbf{t}), k \right\} - \min \left\{ \sum_{j \neq i} \mathbf{1}_{t_j \in E_j} p_j(\mathbf{t}) + p_i(\mathbf{t}), k \right\} d\mathbf{t} \\ & \leq 0 \end{aligned}$$

It can be verified that the left-hand side of this expression is supermodular and symmetric as a function of E . It therefore satisfies the conditions of Lemma 13, and it is enough to check the inequality above for symmetric sets $E_i = E_j$ for all i, j . $\square$

To prove feasibility of the induced interim rules from the merit-with-guarantee mechanism we rely on the following lemma below regarding the existence of a symmetric maximizer. We state and prove the lemma below. To state the result we need to introduce a few pieces of notation.

We say that the disjoint union $D_1 \sqcup \dots \sqcup D_n$ is *symmetric* if $D_i = D_j$ for all i, j . Let H be a set. A function $f : 2^H \rightarrow \mathbb{R}$ is *symmetric* if for all $D_1 \sqcup \dots \sqcup D_n$ and i, j ,

$$f(D_1 \sqcup \dots \sqcup D_i \sqcup \dots \sqcup D_j \sqcup \dots \sqcup D_n) = f(D_1 \sqcup \dots \sqcup D_j \sqcup \dots \sqcup D_i \sqcup \dots \sqcup D_n).$$

Lemma 13. *Let $H := \bigsqcup_i [0, 1]$ and $\mathcal{B}(H)$ denote the subsets $H_1 \sqcup \dots \sqcup H_n$ of H such that H_i is a Borel subset of $[0, 1]$ for each i . Let $f : \mathcal{B}(H) \rightarrow \mathbb{R}$ be supermodular¹³ and symmetric. Consider $G \subseteq \mathcal{B}(H)$ that is closed under permutations, unions, and intersections.*

If $D_1 \sqcup \dots \sqcup D_n \in G$ maximizes f on $\mathcal{B}(H)$ then there is $D' \sqcup D' \sqcup \dots \sqcup D' \in G$ that also maximizes f . In particular, there is a symmetric maximizer.

Proof. Let $D_1 \sqcup \dots \sqcup D_n \in G$ be a maximizer of f . Either

$$f((D_1 \cap D_2) \sqcup (D_1 \cap D_2) \sqcup D_3 \sqcup \dots \sqcup D_n) \geq f(D_1 \sqcup D_2 \sqcup D_3 \sqcup \dots \sqcup D_n)$$

or

$$\begin{aligned} f((D_1 \cap D_2) \sqcup (D_1 \cap D_2) \sqcup D_3 \sqcup \dots \sqcup D_n) &< f(D_1 \sqcup D_2 \sqcup D_3 \sqcup \dots \sqcup D_n) \\ &= f(D_2 \sqcup D_1 \sqcup D_3 \sqcup \dots \sqcup D_n) \end{aligned}$$

where the equality uses symmetry of f .

In the latter case, by denoting $X = (D_1 \cap D_2) \sqcup (D_1 \cap D_2) \sqcup D_3 \sqcup \dots \sqcup D_n$, $Y = D_1 \sqcup D_2 \sqcup D_3 \sqcup \dots \sqcup D_n$ and $Z = (D_2 \setminus D_1) \sqcup (D_1 \setminus D_2) \sqcup \emptyset \sqcup \dots \sqcup \emptyset$, we obtain $f(X) < f(Y)$. By supermodularity, $f(X \cup Z) - f(X) \leq f(Y \cup Z) - f(Y)$, which together with $f(X) < f(Y)$ and symmetry of f implies

$$f(Y) = f(X \cup Z) \leq f(Y \cup Z) = f((D_2 \cup D_1) \sqcup (D_1 \cup D_2) \sqcup D_3 \dots \sqcup D_n).$$

Hence,

$$f(D_1 \sqcup D_2 \sqcup D_3 \sqcup \dots \sqcup D_n) \leq f((D_2 \cup D_1) \sqcup (D_1 \cup D_2) \sqcup D_3 \sqcup \dots \sqcup D_n).$$

Since G is closed under permutations, intersections, and unions, $(D_1 \cap D_2) \sqcup (D_1 \cap D_2) \sqcup D_3 \sqcup \dots \sqcup D_n \in G$ and $(D_2 \cup D_1) \sqcup (D_1 \cup D_2) \sqcup D_3 \sqcup \dots \sqcup D_n \in G$. Repeatedly applying this argument yields the conclusion. $\square$

¹³A function $f : \mathcal{B}(H) \rightarrow \mathbb{R}$ is supermodular if $X, Y, Z \in \mathcal{B}(H)$, $X \subseteq Y$, and $Z \subseteq H \setminus Y$ imply

$$f(X \cup Z) - f(X) \leq f(Y \cup Z) - f(Y).$$

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